Evaluating the Effects of Transport Infrastructure Policy on European Subnational Regions in a Framework with Trade and Commuting

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Abstract

This paper quantifies the regional effects of the European Union's transport infrastructure policy using a structural framework of bilateral trade and commuting across subnational regions in Europe. The model features workers' choice of residence and workplace, and allows both commuting and goods trade to cross national borders and use different transport modes. I construct novel data on bilateral regional trade and commuting flows and on completed road and rail infrastructure projects under the Trans-European Transport Network (TEN-T) between 2010 and 2018. Taking the model to the data, I first estimate the effect of infrastructure project completions on regional bilateral transport costs by mode, finding a robust cost reduction for rail but not for road. Second, I use the estimated rail cost reduction to conduct a counterfactual general equilibrium analysis of the regional effects of the TEN-T pan-European infrastructure policy. The results show that labour mobility accounts for a larger share of the welfare effects than trade, suggesting that large-scale transport infrastructure policy must be evaluated through its effects on both goods and labour mobility to capture the full economic returns.

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1 Introduction

Transport infrastructure shapes where goods are produced and traded (Donaldson and Hornbeck, 2016; Donaldson, 2018), along which routes and modes they are transported (Allen and Arkolakis, 2022; Fuchs and Wong, 2026), and where workers choose to live and work (Baum-Snow, 2007; Heblich, Redding, and Sturm, 2020). Roads and railways are the two central transport modes for moving goods and people in Europe.[1] As the dominant modes, road and rail networks determine the costs of goods and labour mobility between regions and jointly shape the spatial distribution of economic activity. Understanding the effects of large-scale transport infrastructure investment therefore requires a framework in which both trade and commuting can shift between these transport modes.

The European Union (EU) has made transport infrastructure a policy priority through the Trans-European Transport Network (TEN-T), which aims to build a coherent, multimodal transport network for moving goods and people across Europe. Beyond improving overall EU market integration, TEN-T targets regional cohesion by connecting peripheral and lagging regions to the European core and strengthening cross-border links (European Commission, 2013). Since 2014, the policy has concentrated investment on a set of projects under the 2013 and 2024 TEN-T Regulations, with rail receiving the largest share, ahead of road and other modes (European Commission, 2022). Given the scale and ambition of the TEN-T policy, evaluating its economic returns is essential to inform future investment decisions. However, the literature offers little guidance on the combined effects of continent-wide, multimodal transport policy on trade and commuting simultaneously.

While previous studies document the importance of transport infrastructure for trade (e.g. Donaldson, 2018) and commuting (e.g. Ahlfeldt, Redding, Sturm, and Wolf, 2015), they typically study trade or commuting in isolation and focus on a single city or country. A few studies combine trade and commuting (e.g. Monte, Redding, and Rossi-Hansberg, 2018), but also focus on a single country and abstract

[1] See International Transport Forum (2022) for goods and Eurostat (2026) for people.

from mode-specific transport costs, limiting their applicability to a policy like TEN-T, which encompasses multiple countries and targets different transport modes. This paper instead takes a subnational, pan-European perspective and jointly accounts for goods and worker mobility via road and rail. It asks the following question: How important is it to account for both goods mobility and worker mobility when evaluating the welfare and spatial effects of large-scale transport infrastructure investments in road and rail?

To answer this question, I develop a quantitative spatial framework in which subnational regions across the EU trade goods and workers commute between residence and workplace, where trade and commuting can use road and rail. The model delivers general equilibrium effects for trade and commuting flows, wages, prices, and welfare, following a change in transport costs of any mode. I apply the framework to rail and road infrastructure projects under the 2013 TEN-T Regulation.

I construct a novel dataset for 271 regions in 27 European countries by combining three sources: subnational bilateral goods trade flows by transport mode, subnational bilateral commuting flows, and the geo-referenced TEN-T road and rail network. Using the transport network data, I compute the optimal route between every pair of regions for rail and road separately, and identify which routes are affected by a TEN-T infrastructure project. The dataset documents patterns in trade and commuting which motivate key features of my framework. Across region pairs, land transport modes dominate trade flows, with 44% of trade value shipped via road and 31% via rail. Both modes thus carry a substantial and comparable share of traded goods. In 48% of region pairs that trade via land transport, goods are shipped via both road and rail, indicating that the two modes are commonly combined on the same connections. The regional bilateral commuting data shows that 7% of workers, about 15 million people, work in a different NUTS2 region than they reside. This sizeable group of interregional commuters can be directly affected by changes in commuting costs induced by transport infrastructure investment.

To quantify the effects of TEN-T infrastructure policy under the 2013 Regulation, I proceed in two steps. First, I estimate the model's trade gravity equation, exploiting region pairs' exposure to completed TEN-T projects, and recover the associated transport cost reduction. Second, I use the estimated cost reduction as a shock in the structural model to conduct a counterfactual analysis. Starting from the observed equilibrium prior to the policy implementation and changing only the TEN-T-induced transport costs for trade and commuting, I quantify the resulting general equilibrium effects on trade and commuting, wages and prices, as well as welfare.

My structural estimation results suggest there is a robust reduction in transport costs from rail project completions, of about 20%, while road project completions show no robust effect, consistent with an already well-developed road network at the outset. I apply this estimated cost reduction to trade and commuting costs for rail in the general equilibrium analysis. The results reveal substantial heterogeneity across and within countries. In some regions, rail exports increase by more than 100% and rail commuting increases by up to 200%, with corresponding reallocations in residence and workplace patterns. Aggregate welfare rises by 0.13%. Decomposing the aggregate welfare gain, I find that commuting explains the larger share of the welfare effect (62%), compared to goods trade (38%). Labour mobility therefore plays a central role in the welfare effects of EU rail transport infrastructure. Evaluating infrastructure policy through trade alone would substantially understate its returns.

The remainder of this paper is structured as follows. The next section reviews the related literature. Section 3 introduces the TEN-T policy in more detail. Section 4 develops the theoretical model. Section 5 describes the data and Section 6 presents the estimation. Section 7 presents the counterfactual analysis and Section 8 concludes.

2 Literature

This paper builds on a growing body of literature in quantitative spatial economics, which has turned to granular subnational settings to uncover regional heterogeneity in trade and labour mobility in the context of transport infrastructure.[2] My contribution lies at the intersection of several strands of this literature.

One strand quantifies how domestic transport infrastructure affects trade and the distribution of economic activity, building on the seminal contributions of Allen and Arkolakis (2014) and Donaldson (2018), typically within a single country. With applications to the United States, frameworks have been developed and extended to account for routing and congestion (Allen and Arkolakis, 2022; Allen, Fuchs, and Wong, 2025). Further studies consider subnational settings in other countries, including China, Turkey, and India (Faber, 2014; Coşar, Demir, Ghose, and Young, 2022; Baragwanath, Hanson, Khandelwal, Liu, and Park, 2026). Domestic infrastructure improvements have also been studied in settings with international trade (Fajgelbaum and Schaal, 2020; Jaworski, Kitchens, and Nigai, 2023; Bonadio, 2024).

A second strand studies how transport infrastructure affects commuting and the structure of cities, mostly focussing on a single city or country. Ahlfeldt, Redding, Sturm, and Wolf (2015) develop the workhorse quantitative urban model with commuting, applied to the division and reunification of Berlin, and Heblich, Redding, and Sturm (2020) develop a dynamic framework, applied to the expansion of London's Victorian railway network. Further extensions include Allen and Arkolakis (2022, urban variant); Severen (2023); and Dingel and Tintelnot (2026), all applied to US cities.

Only few studies combine trade and commuting in one framework. Monte, Redding, and Rossi-Hansberg (2018) develop the seminal framework linking trade and commuting to show how commuting shapes the response of US local labour markets

[2]See surveys in Redding and Turner (2015); Redding and Rossi-Hansberg (2017); Allen and Arkolakis (2025); Redding (2025); Donaldson (2025).

to economic shocks and to quantify the welfare gains from commuting cost reductions. Krebs and Pflüger (2023) apply the framework to Germany and highlight the role of housing markets for local labour market effects. Recent working papers also take up the combination of trade and commuting. Velasquez (2024) quantifies the effect of the US Interstate Highway System on suburbanisation via trade and commuting cost reductions, and Cosentino (2026) studies the effect of Paris' nineteenth-century transit railroad on city structure and welfare. While these studies focus on one city or country and do not combine multiple transport modes, I combine both trade and commuting with choices of transport mode and apply my framework to a subnational, pan-European setting in which goods and people can cross national borders.

Transport mode choice has been introduced separately into frameworks of trade and of commuting. For trade, Hummels and Schaur (2013) structurally estimate exporters' choice between air and ocean cargo, while Lugovskyy, Skiba, and Terner (2025) model substitution between maritime, air, and land transport to assess a maritime carbon cap. In my framework, freight transport mode choice is similar to Fuchs and Wong (2026), who embed mode choice within a routing model of the US multimodal freight network, with separate elasticities across routes and modes, and apply it to assess the value of the US rail network. Similarly, Tolva (2026) combines discrete mode choice with a Ricardian trade model to study to a closure of Russian airspace and a Suez Canal blockage. Fan, Lu, and Luo (2023) and Mau, Xu, and Zheng (2025) study how exporters choose transport modes and routes in China. For commuting, my framework builds on Herzog (2024), who introduces commuting mode choice for driving and public transit as a nested discrete choice into residence-workplace choices, applied to study London's congestion tax. Similar models have been developed and applied to cities in India, China, the US, and Colombia (Chen, Hasan, Jiang, and Parkhomenko, 2024; Barwick, Li, Waxman, Wu, and Xia, 2024; Almagro, Barbieri, Castillo, and Salz, 2025; Tsivanidis, 2026).

My contribution lies in allowing for a choice among transport modes for goods trade and for commuting. This enables me to quantify how infrastructure investment in specific modes, especially rail, affects the movement of both goods and people, alongside general equilibrium effects.

Turning to the European context, which is the focus of my application, Bröcker, Korzhenevych, and Schürmann (2010) provide an early spatial computable general equilibrium assessment of the welfare effects of the TEN-T network, focusing on goods trade. More recently, Persyn, Barbero, Diaz-Lanchas, Lecca, Mandras, and Salotti (2023) and Shevtsova, Díaz-Lanchas, Persyn, and Mandras (2025) evaluate the general equilibrium effects of EU Cohesion Policy road infrastructure investment, using a sectoral dynamic spatial equilibrium trade model and a structural gravity trade model, respectively. Most of the remaining literature on European transport infrastructure relies on reduced-form or gravity estimation, such as Goldmann and Wessel (2020), who show that newly built TEN-T roads raise regional GDP growth in Eastern European NUTS3 regions. Kaliske (2024) estimates a gravity equation of the effect of TEN-T road construction on regional trade and Ignatov (2024) estimates how the expansion of European highways affects transport costs and regional income. Considering transport modes, Boehm, Arnz, and Winter (2021) study the conditions under which high-speed rail freight could shift European goods transport from road to rail. These studies, structural and reduced-form alike, focus on goods trade and do not model labour mobility.

Commuting has also been studied in the European context. Using a gravity framework, Persyn and Torfs (2016) estimate regional border effects on Belgian inter-municipal commuting. Cross-border labour mobility in Europe and its determinants have been analysed empirically by several studies (Marvakov and Thomas, 2007; Parenti and Tealdi, 2021; Broersma, Edzes, and van Dijk, 2022), some highlighting the role of transport infrastructure (Cavallaro and Dianin, 2019; Loumeau, 2023; Bertram, Chilla, and Hippe, 2024). Considering transport modes, Heuermann and

Schmieder (2019) find that the introduction of high-speed trains led to a modal shift toward rail commuting in Germany, in particular on medium distances.

This literature in the European context leaves two gaps my paper addresses. The few structural general equilibrium approaches focus on trade and do not allow commuting to respond to changes in mode-specific transport costs. The remaining, largely reduced-form literature does not quantify the full general equilibrium effects of infrastructure investment. I combine both goods trade and commuting, each with an explicit choice of transport mode, in a structural general equilibrium framework. In my multi-country, multi-subnational-region setting, goods and people can cross national borders. Hence, my framework is suitable to quantify the general equilibrium effects of TEN-T, an EU policy which coordinates infrastructure investment for different transport modes across countries to efficiently move goods and people throughout Europe.

3 Background

The TEN-T policy is the EU's instrument for developing a coherent, efficient, multimodal transport infrastructure network across EU member states. The policy aims at facilitating the movement of people and goods, deepening the EU's internal market, and strengthening economic and social cohesion across all European regions.[3] The TEN-T consists of a core network, complemented by a comprehensive network, with the goal that by 2050 most of Europe's citizens and businesses can reach it within 30 minutes travel time. The network comprises railways, roads, inland waterways, ports, airports, freight terminals, and urban nodes. To facilitate seamless transportation across the EU, cross-border links and intermodal connections are

[3]Information on the TEN-T can be found, for example, in European Commission (2013), European Parliament (2016), and at https://transport.ec.europa.eu/transport-themes/infrastructure-and-investment/trans-european-transport-network-ten-t_en. An interactive map viewer is available at https://webgate.ec.europa.eu/tentec-maps/web/public/screen/home. Information on TEN-T financing can be found, for example, in European Commission (2022) and at https://www.europarl.europa.eu/factsheets/en/sheet/136/financing-the-trans-european-networks.

promoted. Rail infrastructure takes a prominent position within the TEN-T framework because of its capacity to efficiently transport high volumes of passengers and freight while reducing the environmental and climate impact of transport.

The legal framework of TEN-T is based on EU regulations defining the network's objectives, governance, and financing. Regulation (EU) No 1315/2013 establishes the legal framework for the TEN-T, defining the objectives, priorities, and technical standards for the development of a unified European transport network. It introduces the core network and comprehensive network, sets out implementation requirements, and concentrates resources on projects of European relevance. Regulation (EU) 2024/1679 updates the TEN-T framework, revising the network structure, implementation requirements, and development priorities for the TEN-T, with the objective of improving the sustainable, resilient, and interoperable movement of people and goods across the European Union. Funding of the TEN-T was established in Council Regulation (EC) No 2236/95 (1995), which was superseded by further regulations, such as regulations on the Connecting Europe Facility (CEF), which has been the main EU funding instrument for TEN-T investments since 2013 (see, e.g. Regulation (EU) 2021/1153, establishing the CEF II 2021-2027).

TEN-T policy is implemented through individual infrastructure projects that involve the construction of new transport links, the upgrading of existing infrastructure, the removal of bottlenecks, and improvements in interoperability. Particular emphasis is placed on missing links and cross-border projects with a European added value. The planning and implementation of TEN-T projects involve cooperation between the European Commission, the European Climate, Infrastructure and Environment Executive Agency (CINEA), member states, and regional authorities.

The projects are financed through different sources, where the EU contributes mainly through the CEF, complemented by contributions from the Structural and Cohesion Funds. For implementation of the network set out in the 2013 Regulation (1315/2013), the European Union allocated 23.2 billion euros to transport

investments through the CEF for the period 2014-2020. Rail projects constitute a large share of investment, with about 70% of the overall budget. For the period 2014-2020 this amounted to 16.24 billion euros for 448 rail projects. Road projects receive a substantially smaller share, with CEF funding of about 2 billion euros for 223 projects in the period 2014-2020. For the ongoing period 2021-2027, the budget allocated to CEF amounts to 25.8 billion euros, among which about 18 billion euros will be spent on rail projects. EU funding through the CEF is intended to act as a catalyst for projects, while most of the funding comes from national, regional, and private sources, as well as international financial institutions such as the European Investment Bank. Generally, the EU supports TEN-T infrastructure projects with grants covering up to 30% of the costs, whereas cross-border railway projects may receive up to 50% and projects in peripheral regions may receive up to 70% (see Marketa, 2021).

Having access to trade and commuting data for 2010-2018, my study focuses on the projects conducted under the 2013 Regulation, using geospatial data on the locations of the projects and researched data on the time of completion. My estimates of the reduction in transport costs for these projects allows to conduct different counterfactual scenarios. While my analysis focuses on the general equilibrium effects of the projects conducted under the 2013 Regulation, it can equally be applied to analyse the effects of current projects from the 2024 Regulation, or future projects.

4 Theoretical Model

In the multi-country, multi-region model of trade and commuting with transport mode choices, each country consists of a set of subnational regions. Regions are indexed by k, n, and r, and the total set of regions in all countries is denoted by $\mathcal{R}$. Goods are differentiated by region of origin, following Armington (1969). Mobile workers choose where to live and work, as in Monte, Redding, and Rossi-

Hansberg (2018), and, facing commuting costs, they decide via which transport mode to commute, as in Herzog (2024). They also choose where to source their consumption goods from, as in Monte, Redding, and Rossi-Hansberg (2018), and via which transport mode, as in Fuchs and Wong (2026). In the economy, labour is the only factor of production and one unit of labour produces one unit of output. Assuming perfect competition, the factory gate price of the goods produced in region n equals the wage in the same region, $p_n = w_n$.

An individual worker ω resides in region k and inelastically supplies her unit of labour in her workplace region n, where n is either in the same country as k or in a neighbouring country. She commutes from her residence to her workplace via mode $m \in \mathcal{M}$ and allocates her spending on consumption goods from each origin region r across available transport modes $l \in \mathcal{M}$.[4] Worker ω is characterised by the following utility function:

$$U_{kn}^{m}(\omega) = \left[\sum_{r \in \mathcal{R}} \left(\left[\sum_{l \in \mathcal{M}} \left(c_{rk}^{l} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \frac{b_{kn}(\omega) v^{m}(\omega)}{d_{kn}^{m}}, \tag{1}$$

where the first term is a nested constant elasticity of substitution (CES) consumption aggregator over goods originating from different regions and shipped via different transport modes, and the second term captures worker-specific preferences over residence-workplace locations and commuting modes, formalised as a nested discrete choice.

In the consumption aggregator, $\sigma > 1$ is the elasticity of substitution between goods from different regions and $\rho > 1$ is the elasticity of substitution between freight transport modes. c_{rk}^{l} denotes the quantity of the good produced in origin region r, consumed in region k, and shipped via mode l. The CES demand structure over regional varieties reflects imperfect substitutability and love of variety across

[4] The set of transport modes $\mathcal{M}$ is assumed to be the same for both shipping goods and commuting, as the application focuses on the road and rail network of the TEN-T.

goods originating from different regions. Product differentiation across transport modes for a regional variety may arise because modes differ in characteristics such as delivery speed, the preservation of product quality, and environmental image. Goods shipped via different transport modes can therefore be treated as differentiated varieties and imperfect substitutes in the CES nest over transport modes.[5] In the location choice part, $b_{kn}(\omega)$ are worker-specific residence-workplace preference shocks, independently drawn from a Fréchet distribution. $d_{kn}^m \geq 1$ are iceberg-type costs of commuting, measured in units of utility, between residence k and workplace n via mode m.[6] $v^m(\omega)$ are worker-specific commuting-mode preference shocks, independently drawn from a Fréchet distribution.

The worker earns her wage w_n in workplace region n, which she spends on consumption goods in residence region k, given the budget constraint[7]

$$w_n = \sum_{r \in \mathcal{R}} c_{rk} p_{rk}, \tag{2}$$

where c_{rk} is the consumption index over goods originating from r consumed by the

[5]Both delivery speed and environmental characteristics have been shown to be relevant for consumer demand. Hummels and Schaur (2013) model consumers as perceiving slower-delivered goods as lower quality, rationalising the choice between air and ocean freight. On externalities attached to product characteristics, Krautheim and Verdier (2016) model consumer demand for goods produced with "clean" technology. Hainmueller, Hiscox, and Sequeira (2015) find experimental evidence that US consumers value ethical labelling as a product attribute. Since transport modes differ in the associated emissions, such preferences can apply to mode-specific demand. Lugovskyy, Skiba, and Terner (2025) and Cristoforoni, Errico, Rodari, and Tolva (2026) similarly allow for consumer-side transport mode choice. Therefore, the model description uses the nested CES structure for workers' choices over origin and shipping mode of consumption goods.
However, transport modes may only be transparent to consumers in some purchasing decisions, such as online retail. In addressing this, note that the same trade shares by transport mode are obtained under two further, isomorphic microfoundations. A second, isomorphic microfoundation models transport mode choice as a (consumer's) discrete choice over modes, as in Tolva (2026) and Fuchs and Wong (2026). I show the equivalence in Appendix B. In a third microfoundation, a retailer assembling the consumption bundle draws a mode-specific cost shock per shipment and chooses the cheapest mode. This last microfoundation requires no consumer awareness of mode at all. All three microfoundations are isomorphic and deliver identical trade shares by transport mode, which is what the application requires and what I observe in the data.

[6]Iceberg commuting costs are often used in commuting models. For example, Ahlfeldt, Redding, Sturm, and Wolf (2015) note that iceberg commuting costs reduce the effective units of labour, as they enter the indirect utility function multiplicatively, which is also the case here, in Equation (6) below.

[7]The assumption that workers spend their entire income on goods in the residence region is in line with existing studies, such as Monte, Redding, and Rossi-Hansberg (2018).

worker in k, aggregated over all transport modes, $c_{rk} = \left[\sum_{l \in \mathcal{M}} \left(c_{rk}^{l}\right)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}$, and p_{rk} is the price index of goods produced in r that are consumed in k, including transport costs.[8]

The worker makes four choices. First, she chooses a residence-workplace pair and a mode of transportation for commuting. Given the location decision, the worker makes her consumption choices, in which she decides where to source her consumption goods from. For every origin region, she chooses how much to ship via each transport mode to her region of residence. In the following sections, I first derive the demand for goods for given residence-workplace locations, before determining location and commuting choices. More details on the model and derivations are given in Appendix A.

4.1 Demand for Goods

Conditional on places of residence and work, the worker makes her consumption choices by deciding where to purchase her consumption goods and via which transport mode they are shipped.

Solving the maximisation problem for the outer CES nest over regional varieties, I obtain the value of the worker's demand for consumption goods from region r:

$$c_{rk} p_r t_{rk} = \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma} w_n, \tag{3}$$

where the price of delivered goods is the factory-gate price in the origin region multiplied with transport costs, $p_{rk} = p_r t_{rk}$. Transport costs are an index over mode-specific iceberg transport costs, specified below. P_k is the price aggregate for goods consumed in region k, with $P_k^{1-\sigma} = \sum_{r' \in \mathcal{R}} \left(p_{r'k}\right)^{1-\sigma}$.

For each origin region, the worker decides on the quantity of consumption goods

[8] As shown below, transport costs from region r to region k are an index over the mode-specific transport costs for all shipping modes. The delivered price p_{rk} hence is a price index of goods from region r consumed in region k, aggregating the transport costs for all shipping modes.

shipped via each transport mode. Solving the maximisation problem for the inner CES nest over transport modes, the value of the worker's demand for goods from region r shipped via mode l is

$$c_{rk}^{l} p_r t_{rk}^{l} = \frac{\left(p_r t_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(p_r t_{rk}^{l'}\right)^{1-\rho}} c_{rk} p_{rk}, \tag{4}$$

where the price of delivered goods shipped via mode l is $p_{rk}^{l} = p_r t_{rk}^{l}$, with mode-specific iceberg transport costs $t_{rk}^{l} \geq 1$. Defining the price index for goods from r consumed in k as $p_{rk}^{1-\rho} = \sum_{l \in \mathcal{M}} \left(p_{rk}^{l}\right)^{1-\rho}$, I obtain

$$p_{rk} = p_r \left[\sum_{l \in \mathcal{M}} \left(t_{rk}^{l}\right)^{1-\rho} \right]^{\frac{1}{1-\rho}},$$

where the transport cost index is defined by

$$t_{rk} = \left[\sum_{l \in \mathcal{M}} \left(t_{rk}^{l}\right)^{1-\rho} \right]^{\frac{1}{1-\rho}}. \tag{5}$$

The aggregate price index for goods consumed in region k can hence be expressed by

$$P_k^{1-\sigma} = \sum_{r \in \mathcal{R}} \left(p_r \left[\sum_{l \in \mathcal{M}} \left(t_{rk}^{l}\right)^{1-\rho} \right]^{\frac{1}{1-\rho}} \right)^{1-\sigma}.$$

4.2 Location and Commuting Choices

Worker mobility is modelled as a nested discrete choice problem. In the upper nest workers choose a residence-workplace pair, while in the lower nest they choose the transport mode for commuting between the locations. For their choice of a residence-workplace pair, workers draw $b_{kn}(\omega)$ independently from a Fréchet distribution with shape parameter $\theta > 1$ (the location choice elasticity). After choosing a residence-workplace pair, workers draw $v^{m}(\omega)$ independently from a Fréchet dis-

tribution with shape parameter $\eta > 1$ (the commuting mode choice elasticity) and choose a transport mode for commuting from their residence to their workplace.

Commuting Mode Choice

Conditional on a residence-workplace pair, workers choose a mode of transportation via which they commute between the two. They choose commuting mode m if the utility obtained using mode m is at least as high as the utility obtained using any other mode m' among the set of transport modes $\mathcal{M}$. Indirect utility is given by:

$$U_{kn}^{m*}(\omega) = \frac{w_n}{P_k} \frac{b_{kn}(\omega) v^m(\omega)}{d_{kn}^m}. \tag{6}$$

The probability that a worker chooses mode m, conditional on having chosen to commute from k to n, is[9]

$$Prob_{m|kn} = Prob\left(\frac{w_n}{P_k} \frac{b_{kn} v^m}{d_{kn}^m} \geq \frac{w_n}{P_k} \frac{b_{kn} v^{m'}}{d_{kn}^{m'}} \quad \forall \quad m' \neq m \right)$$

I derive the choice probability as in Herzog (2024) and according to Train (2009) and obtain the probability of a worker choosing commuting mode m, given that she commutes from k to n:[10]

$$Prob_{m|kn} = \frac{(d_{kn}^m)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'}\right)^{-\eta}}. \tag{7}$$

The probability that a worker chooses mode m is higher, the lower the commuting

[9] For ease of notation, I am omitting ω from the expressions. So, in the following, v^m is used to denote $v^m(\omega)$ and b_{kn} is used to denote $b_{kn}(\omega)$.

[10] Note that assuming Fréchet distributed multiplicative idiosyncratic shocks does not lead exactly to the multinomial logit choice probability derivation following McFadden (1974), which uses additive idiosyncratic shocks that are Gumbel distributed. The obtained choice probabilities (or shares) are similar in both cases, as also noted in Monte, Redding, and Rossi-Hansberg (2018, footnote 12) and Allen and Arkolakis (2014, p. 1120). Many studies use Fréchet distributed idiosyncratic shocks in discrete location choice models, such as Ahlfeldt, Redding, Sturm, and Wolf (2015); Monte, Redding, and Rossi-Hansberg (2018); Allen and Arkolakis (2022); Herzog (2024), while other studies, such as Dingel and Tintelnot (2026) use the Gumbel distribution. Details can also be found in Mattsson, Weibull, and Lindberg (2014).

cost of mode m relative to the other available modes, with η determining how sensitive the choice probability is to cost differences between modes.

Location Choice

Workers decide where to live and work, before choosing a mode of transportation for their commute. To exclude unrealistically far commutes from the residence k, the workplace n can only be chosen among the set of regions in the country of residence or in a neighbouring country, denoted by $\mathcal{R}(k) \subseteq \mathcal{R}$. Note that the admissibility relation between possible residence and workplace regions is symmetric, such that $k \in \mathcal{R}(n)$ whenever $n \in \mathcal{R}(k)$. Location choice is based on workers' location preference shocks b_{kn} and expected utility. Ex ante expected mean utility of commuting from k to n replaces $\frac{v^m}{d_{kn}^m}$ in the indirect utility expression with the inverse of a commute cost index $\overline{d_{kn}}$:[11]

$$\mu_{kn} = \frac{w_n}{P_k \overline{d_{kn}}} b_{kn}.$$

A worker chooses to live in region k and work in region n if the utility obtained from this location pair is at least as high as the utility obtained from any other residence-workplace pair among the set of all other admissible region pairs. The probability that worker ω chooses residence-workplace pair kn, where $n \in \mathcal{R}(k)$, is

$$Prob_{kn} = Prob\left(\frac{w_n}{P_k \overline{d_{kn}}} b_{kn} \geq \frac{w_s}{P_r \overline{d_{rs}}} b_{rs} \quad \forall \quad rs \neq kn, \text{ with } r \in \mathcal{R},\, s \in \mathcal{R}(r)\right).$$

Deriving the choice probability analogously to the commuting mode choice, I obtain

[11] $\frac{1}{\overline{d_{kn}}}$ is defined as the expected value of the maximum of $\frac{v^m}{d_{kn}^m}$ with respect to commuting transport mode m. $\frac{1}{\overline{d_{kn}}}$ is analogous to what is referred to as the inclusive value in nested logit choice models, corresponding to the expected value of the utility a worker obtains from the alternatives among the set of transport modes for commuting between a given residence-workplace pair. It is specified in detail in Appendix A.5, in line with Herzog (2024).

the probability of a worker choosing to live in k and work in n:

$$Prob_{kn} = \frac{\left(\frac{w_n}{P_k \bar{d}_{kn}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \bar{d}_{rs}}\right)^{\theta}}. \tag{8}$$

The probability that a worker chooses to live in region k and work in region n is higher, the higher the real wage realised by the residence-workplace pair and the lower the expected commuting cost between k and n, relative to all other residence-workplace pairs. θ determines how sensitive the choice probability is to differences in real wages net of commuting costs between region pairs.

4.3 Regional Bilateral Flows

Trade Flows

To obtain aggregate trade from region r to region k, X_{rk}, recall that commuting separates the regions where workers earn their wages and where they consume goods. Define the number of workers living (denoted by superscript R for residence) in region k as $L_k^R = \sum_{n \in \mathcal{R}(k)} L_{kn}$. Define the number of workers working (denoted by superscript L for labour) in region n as $L_n^L = \sum_{k \in \mathcal{R}(n)} L_{kn}$. Output of region n is the wage in region n multiplied with the total number of workers in n:

$$Y_n = w_n \sum_{k \in \mathcal{R}(n)} L_{kn} = w_n L_n^L.$$

Expenditure of region k is the sum over workplaces of the product of each workplace's wage with the number of its workers who reside in k:

$$E_k = \sum_{n \in \mathcal{R}(k)} w_n L_{kn}.$$

In the presence of regional trade imbalances, expenditure of a region k may include a trade deficit (or surplus), such that expenditure can be expressed by $E_k =$

$\sum_{n\in\mathcal{R}(k)} w_n L_{kn} + D_k$, where D_k is the aggregate regional trade imbalance (deficit or surplus) of region k. Since this framework does not aim at explaining trade imbalances, details on the inclusion of trade imbalances are provided in Appendix A.4. To account for trade imbalances observed in the data, the equation for expenditure in the equilibrium system of equations (22)–(29) below includes the exogenous term D_k. It is used to run a preliminary counterfactual analysis where trade imbalances are eliminated. The resulting values, consistent with zero trade imbalances, are used as baseline values for conducting the counterfactual analysis.

Multiplying the demand for goods from region r of a worker living in region k and working in region n from Equation (3) with L_{kn} gives the demand for goods from region r of all workers living in k and working in n:

$$c_{rk} p_r t_{rk} L_{kn} = \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma} w_n L_{kn}.$$

Summing over all workplaces and substituting expenditure, aggregate trade from region r to region k is given by:

$$X_{rk} = \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma} E_k. \tag{9}$$

Expressing the spending of region k on goods from region r as share of total spending of region k yields

$$\pi_{rk} = \frac{X_{rk}}{E_k} = \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma}. \tag{10}$$

To obtain regional trade by transport mode, I multiply Equation (4), which gives the worker's demand for goods from region r shipped via mode l, with the number of residents in k:

$$c^l_{rk} p_r t^l_{rk} L^R_k = \frac{\left(p_r t^l_{rk}\right)^{1-\rho}}{\sum_{l'\in\mathcal{M}} \left(p_r t^{l'}_{rk}\right)^{1-\rho}} c_{rk} p_r t_{rk} L^R_k.$$

With the factory-gate price being independent of the transport mode used for delivering goods to the regions where they are consumed and using the definition of aggregate regional trade from Equation (9), the aggregate trade shipped via mode l from region r to region k is

$$X_{rk}^{l} = \frac{\left(t_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}} X_{rk}. \tag{11}$$

The share of trade from r to k shipped via mode l in total trade from r to k depends only on transport costs by mode:

$$\frac{X_{rk}^{l}}{X_{rk}} = \frac{\left(t_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}}. \tag{12}$$

Goods market clearing implies that total output of each region is equal to sales to all destination regions,

$$Y_r = \sum_{k \in \mathcal{R}} X_{rk} = \sum_{k \in \mathcal{R}} \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma} E_k. \tag{13}$$

Structural Gravity System: Regional Trade

I derive the structural gravity system of regional trade, following Anderson and van Wincoop (2003), and obtain:

$$X_{rk} = \frac{Y_r E_k}{Y} \left(\frac{t_{rk}}{\Pi_r P_k}\right)^{1-\sigma}, \tag{14}$$

where the outward and inward multilateral resistance terms are given by:

$$\Pi_r^{1-\sigma} = \sum_{k \in \mathcal{R}} \left(\frac{t_{rk}}{P_k}\right)^{1-\sigma} \frac{E_k}{Y} \tag{15}$$

$$P_k^{1-\sigma} = \sum_{r \in \mathcal{R}} \left(\frac{t_{rk}}{\Pi_r}\right)^{1-\sigma} \frac{Y_r}{Y}. \tag{16}$$

The gravity equation of trade by transport mode is given by:

$$X_{rk}^{l} = \frac{Y_r E_k}{Y} \left(\frac{1}{\Pi_r P_k} \right)^{1-\sigma} \frac{\left(t_{rk}^{l}\right)^{1-\rho}}{\left[\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}\right]^{\frac{\sigma-\rho}{1-\rho}}}. \tag{17}$$

Commuting Flows

The probability of a worker choosing to live in k and work in n from Equation (8) coincides with the share of total workers $\overline{L}$ who choose to live in region k and work in region n. Hence, the bilateral regional commuting flow from k to n is given by

$$L_{kn} = \frac{\left(\frac{w_n}{P_k \overline{d}_{kn}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \overline{d}_{rs}}\right)^{\theta}} \overline{L}. \tag{18}$$

Summing the bilateral commuting share across all possible workplaces, I obtain the probability that a worker chooses to live (denoted by superscript R) in region k, which is also the share of the total number of workers across all regions who live in region k:

$$\frac{L_k^R}{\overline{L}} = \sum_{n \in \mathcal{R}(k)} \frac{L_{kn}}{\overline{L}} = \sum_{n \in \mathcal{R}(k)} \frac{\left(\frac{w_n}{P_k \overline{d}_{kn}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \overline{d}_{rs}}\right)^{\theta}}. \tag{19}$$

Summing the bilateral commuting share across all possible residences, I obtain the probability that a worker chooses to work (denoted by superscript L) in region n, which is also the share of the total number of workers across all regions who work in region n:

$$\frac{L_n^L}{\overline{L}} = \sum_{k \in \mathcal{R}(n)} \frac{L_{kn}}{\overline{L}} = \sum_{k \in \mathcal{R}(n)} \frac{\left(\frac{w_n}{P_k \overline{d}_{kn}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \overline{d}_{rs}}\right)^{\theta}}. \tag{20}$$

Turning to the commuting modes, the probability of a worker choosing commut-

ing mode m for her commute from k to n, given in Equation (7), coincides with the share of workers who commute from k to n using mode m. Hence, the regional commuting flow from k to n via mode m, is given by

$$\begin{aligned} L_{kn}^{m} =& \frac{\left(d_{kn}^{m}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'}\right)^{-\eta}} L_{kn} \\ =& \frac{\left(d_{kn}^{m}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'}\right)^{-\eta}} \frac{\left(\frac{w_n}{P_k \overline{d}_{kn}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \overline{d}_{rs}}\right)^{\theta}} \overline{L}. \end{aligned} \tag{21}$$

4.4 General Equilibrium

I follow Dekle, Eaton, and Kortum (2007, 2008) and formulate the structural framework in changes, which has the advantage that trade and commuting costs are not specified in levels, as well as eliminating variables that are constant over time. This implies that the only data necessary to implement a counterfactual analysis of transport infrastructure policy is regional bilateral trade and commuting flows by transport mode, alongside changes in trade and commuting costs implied by the infrastructure policy, as well as parameters $\{\sigma, \rho, \eta, \theta\}$.

Let a hat above a variable denote the variable's counterfactual value relative to its baseline value, e.g. $\widehat{\pi}_{rk} = \frac{\pi_{rk}^{c}}{\pi_{rk}^{b}}$. Taking into account that perfect competition implies wages equal to producer prices $w_n = p_n$ and treating trade imbalances D_k as exogenous, the equilibrium in changes is given by:[12]

$$\widehat{\pi}_{rk} = \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\widehat{P}_k^{1-\sigma}} \tag{22}$$

$$\widehat{Y}_r Y_r^b = \sum_{k \in \mathcal{R}} \widehat{\pi}_{rk} \widehat{E}_k \pi_{rk}^b E_k^b \tag{23}$$

$$\widehat{w}_n = \frac{\widehat{Y}_n}{\sum_{k \in \mathcal{R}(n)} \frac{L_{kn}^b}{\sum_{j \in \mathcal{R}(n)} L_{jn}^b} \widehat{L}_{kn}} \tag{24}$$

[12] Details on the model equations in levels can be found in Appendix A.6. Derivations can be found in Appendix A.7

$$\widehat{E}_k E_k^b = \sum_{n \in \mathcal{R}(k)} \widehat{w}_n \frac{Y_n^b}{\sum_{j \in \mathcal{R}(n)} L_{jn}^b} \widehat{L}_{kn} L_{kn}^b + D_k \tag{25}$$

$$\widehat{P}_k = \left(\sum_{r \in \mathcal{R}} \pi_{rk}^b \left(\widehat{w}_r \widehat{t}_{rk} \right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \tag{26}$$

$$\widehat{t}_{rk} = \left[\sum_{l \in \mathcal{M}} \pi_{rk}^{l,b} \left(\widehat{t}_{rk}^{l} \right)^{1-\rho} \right]^{\frac{1}{1-\rho}} \tag{27}$$

$$\widehat{L}_{kn} = \frac{\left(\frac{\widehat{w}_n}{\widehat{P}_k \widehat{\overline{d_{kn}}}} \right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \frac{L_{rs}^b}{L} \left(\frac{\widehat{w}_s}{\widehat{P}_r \widehat{\overline{d_{rs}}}} \right)^{\theta}} \tag{28}$$

$$\widehat{\overline{d_{kn}}} = \left(\sum_{m \in \mathcal{M}} \frac{L_{kn}^{m,b}}{L_{kn}^b} \left(\widehat{d}_{kn}^{m} \right)^{-\eta} \right)^{-\frac{1}{\eta}} \tag{29}$$

The trade and commuting flows by transport mode are not necessary to solve the system of equations and can be reported after solving the general equilibrium:

$$\widehat{X}_{rk} = \widehat{\pi}_{rk} \widehat{E}_k \tag{30}$$

$$\widehat{X}_{rk}^{l} = \frac{\left(\widehat{t}_{rk}^{l} \right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \pi_{rk}^{l',b} \left(\widehat{t}_{rk}^{l'} \right)^{1-\rho}} \widehat{X}_{rk} \tag{31}$$

$$\widehat{L}_{kn}^{m} = \frac{\left(\widehat{\overline{d_{kn}}} \right)^{\eta}}{\left(\widehat{d}_{kn}^{m} \right)^{\eta}} \widehat{L}_{kn}. \tag{32}$$

Expected Utility

Worker welfare can be expressed by expected utility, which is the expected value of indirect utility, where workers make their optimal consumption, residence-workplace, and commuting-mode choices.[13] Ex ante expected utility is the expected maximum utility of any worker before realising the residence-workplace preference shocks b_{kn}

[13] Baqaee and Burstein (2026) use the term "average" utility rather than "expected" utility, since it is not a von Neumann–Morgenstern expected utility function. While acknowledging the distinction, I stick with the commonly used term of "expected" utility, used also by Herzog (2024) and Tsivanidis (2026).

and the commuting-mode preference shocks v^m and is given by[14]

$$E\left[\tilde{U}^*\right] = \left(\sum_{kn}\left(\frac{w_n}{P_k\overline{d_{kn}}}\right)^{\theta}\right)^{\frac{1}{\theta}}\Gamma\left(1-\frac{1}{\theta}\right) \tag{33}$$

Ex ante expected utility for all workers is the same. Ex post, different workers have different welfare due to different idiosyncratic shocks. Expected utility increases with wages in the workplace region, while it decreases with prices in the residence region and with commuting costs. This means that wage increases and commuting cost reductions raise welfare, while higher living costs reduce welfare. The change in expected utility is given by

$$\widehat{E\left[\tilde{U}^*\right]} = \left[\sum_{kn}\frac{L_{kn}^b}{\overline{L}^b}\left(\frac{\widehat{w}_n}{\widehat{P}_k\widehat{\overline{d_{kn}}}}\right)^{\theta}\right]^{\frac{1}{\theta}}. \tag{34}$$

5 Data

I construct a novel dataset which combines bilateral trade flows and bilateral commuting flows between 271 NUTS2 regions in 27 European countries over the period 2010–2018 (most EU member states, Norway, and Switzerland) with data on infrastructure constructions and improvements along the TEN-T rail and road networks.

5.1 Regional Trade Flows

The data source for regional bilateral trade flows by transport mode is the European Spatial Planning Observation Network (ESPON) Interregional Relations in Europe (IRiE).[15] I use the interregional trade in goods dataset, which provides intra-national and international region-to-region flows of goods by value between NUTS2 regions for the period 2010-2018, distinguishing five transport modes (road,

[14] Details are given in Appendix A.8.

[15] ESPON (2022). Interregional Relations in Europe (IRiE). Retrieved from https://gis-portal.espon.eu/arcgis/apps/sites/#/irie-hub/pages/trade_goods

rail, ship, aircraft, and others). The dataset was constructed by Llano Verduras, Pérez-Balsalobre, and Gallego López (2021), transforming region-to-region freight statistics by mode (see Schwarze, Spiekermann, Llano Verduras, Pérez-Balsalobre, and Gallego López, 2022) from physical units (tons) into monetary units (millions of euros) using estimated unit prices. Exports and imports were harmonised between country pairs using country-level trade datasets (Eurostat COMEXT, CEPII BACI). The dataset also contains regional trade within countries, where intra-regional and inter-regional flows were harmonised with input-output and trade tables (FIGARO).

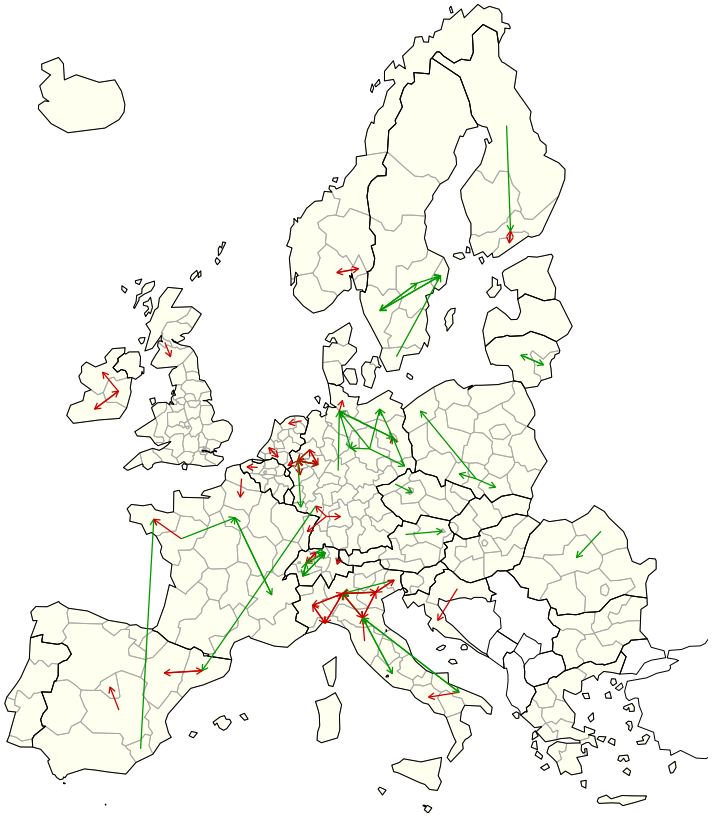
I focus on land transport, that is, road and rail, which carry the majority of goods trade. The average of the modal splits calculated for each region pair and year shows that land transport modes account for 75% of traded goods value, with 44% shipped via road and 31% via rail.

Figure 1 shows the 50 largest trade flows by value in 2010 via rail and road, in Panel (a) among all interregional flows and in Panel (b) among the interregional flows that cross national borders. It shows that road freight is typically used for shorter distances, while the flows by rail cover longer distances. In the international trade flows by road, there are some flows that cannot have been done exclusively by road, without a ferry transit, such as the flow from Eastern and Midland Ireland to East Flanders in Belgium. To exclude unrealistic overland flows, I only keep those regions in my dataset that are connected to the TEN-T road and rail network, explained below.

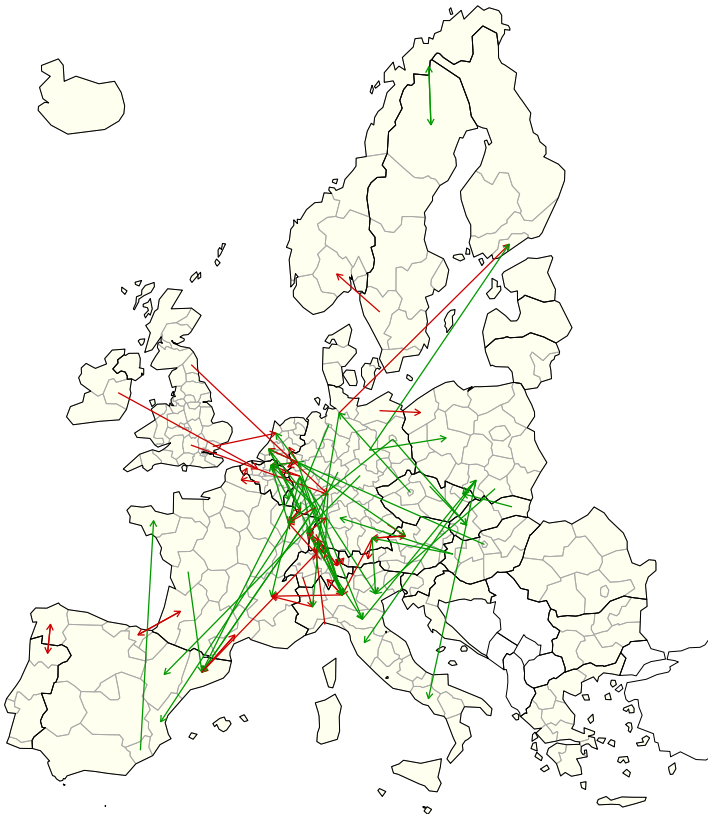
5.2 Regional Commuting Flows

Commuting flows between and within NUTS2 regions for the period 2010-2018 are drawn from the EU Labour Force Survey (LFS). Respondents report their region of residence and region of work. Survey weights, which give the number of individuals in the population each respondent represents, allow me to compute aggregate commuting flows between and within NUTS2 regions. As a consistency check, aggre-

Figure 1: Descriptive Statistics: Trade Flows in 2010.



(a) Top 50 interregional trade flows by value, road (red) and rail (green), 2010.



(b) Top 50 international trade flows by value, road (red) and rail (green), 2010.

gating all employed persons in my data, defined as all those reporting both a region of residence and a region of work, and comparing to regional population statistics, yields an implied employment-to-population ratio of 44.2%, close to the EU's actual employment-to-population ratio of 51.5%.[16] Of the commuting flows, 93.2% are intraregional, 6.1% are interregional within countries, and 0.7% cross national borders.[17]

I do not explicitly model commuting for workers who live and work in the same region. While some who work in their region of residence may live close enough that they do not use rail or road infrastructure at all, others commute a considerable distance within their NUTS2 region. I do not distinguish between the two. I restrict interregional commuting to regions in the same country or in neighbouring countries, to avoid commutes that are unrealistically long to undertake on a daily basis, as described above in Section 4.[18] The group of workers directly exposed to interregional transport improvements therefore consists of interregional commuters across NUTS2 regions within or between neighbouring countries, totalling 14 to 16 million people depending on the year.

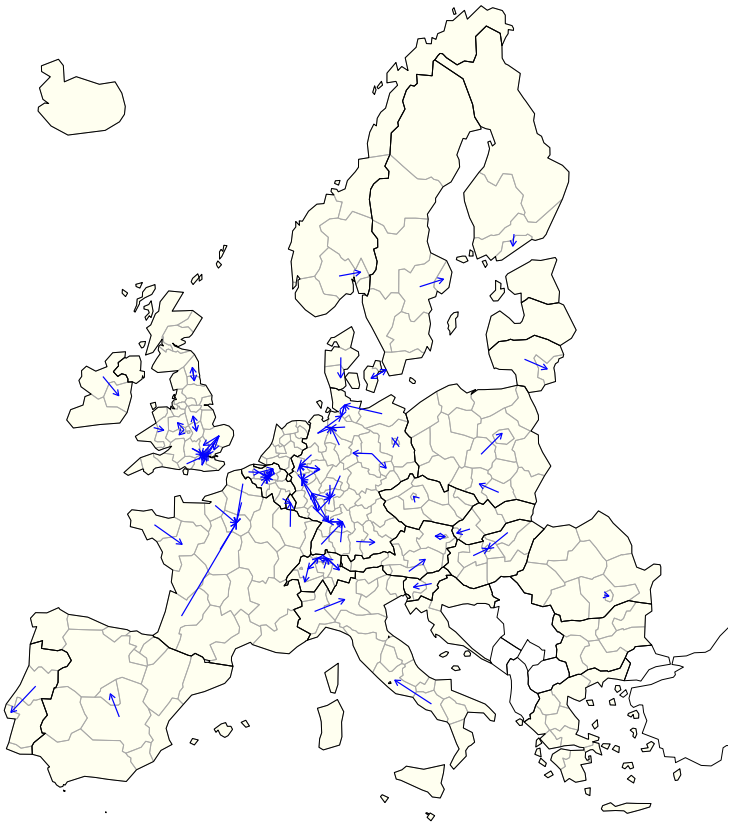
Panel (a) of Figure 2 shows the 100 largest interregional commuting flows in 2010, with strong flows into London, between Belgian and West German regions, and into Paris. Panel (b) shows outcommuters, workers living in a region but working in a different one, as a share of the region's employed population in 2010. Outcommuting shares are highest in Belgium, reaching up to 47%, and also exceed 15% in the South of England, the North of France, Switzerland, and parts of Denmark and Germany.

[16] Population statistics taken from ARDECO, https://territorial.ec.europa.eu/ardeco/viewer/SNPTD. EU employment-to-population ratio taken from the World Bank, https://data.worldbank.org/indicator/SL.EMP.TOTL.SP.NE.ZS?locations=EU.

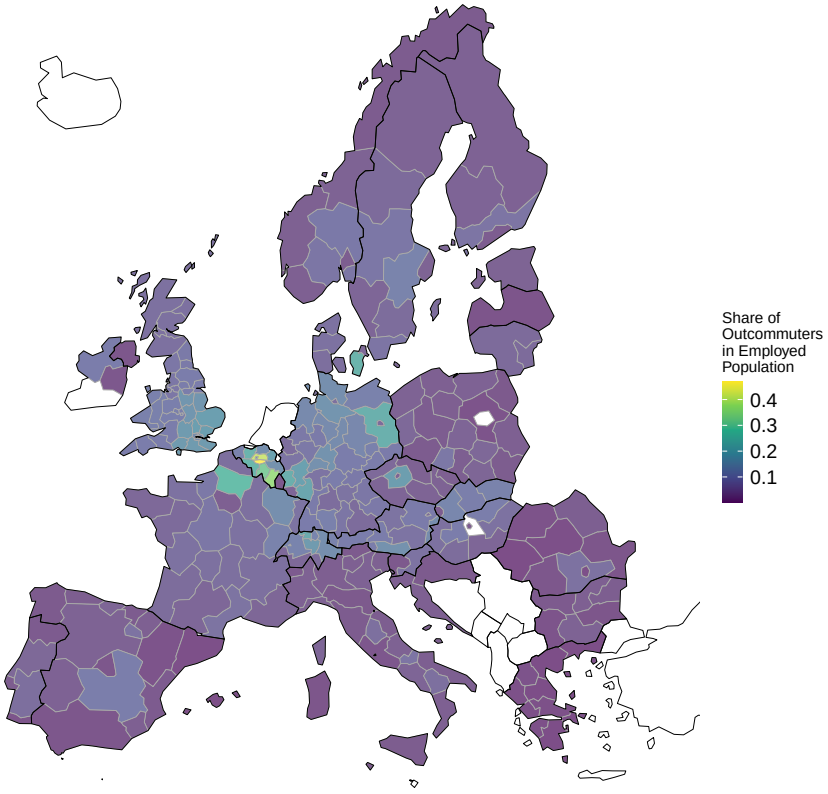
[17] For some international commutes, the exact region of work is not reported. I regionalise these commuting flows across the the regions within the country of work using economic attractiveness (regional GDP) and geographic distance to the region of residence. The LFS does not report region of work for the Netherlands, so all Dutch workers are assumed to work in their region of residence.

[18] This excludes commuters who cross borders on a weekly or working-day basis, which is necessary in a model where workers spend their entire income in their region of residence. The restriction is also consistent with the model. With idiosyncratic discrete-choice shocks, some workers would otherwise choose implausibly long commutes with positive, albeit small, probability if all region pairs were admissible.

Figure 2: Descriptive Statistics: Commuting Flows in 2010.



(a) Top 100 interregional commuting flows, 2010.



(b) Share of outcommuters in the number of employed, 2010.

Unfortunately, the EU LFS does not provide any information on the mode of transportation used to get to work, which is why I add an additional data source to obtain regional commuting flows by transport mode. I use country-level survey information from the Eurobarometer on transport mode use (European Commission Directorate-General for Mobility and Transport, 2024).[19] This implies that the shares of road versus rail use of commuting flows in my data does not vary across time and across regions within the same country. For this reason, I only use the commuting flows to obtain a baseline commuting matrix for my general equilibrium analysis and refrain from using them in estimations of the partial effects of transport infrastructure improvements on commuting flows.

5.3 Trans-European Transport Network

Construction of the EU transport infrastructure data is based on shapefiles of the TEN-T network corresponding to the 2013 and 2024 EU Regulations.[20] The shapefiles contain geodata on the transport network for different transport modes (roads, railway, inland waterways), as well as on nodal infrastructure (airports, ports, railroad terminals, urban nodes). I focus on the road and rail networks, which carry the large majority of European freight and passenger traffic. I identify which regions are connected to the road and rail networks and exclude those that are not (e.g. the French overseas territories, Greek islands, all Irish regions, and Iceland), leaving 271 NUTS2 regions in 27 countries (most EU member states, Norway, and Switzerland) connected by road and rail.

For each bilateral region pair, I compute the optimal route along the road network and, separately, along the rail network. Each region is represented by the network

[19] The question asked is: "In the past two years, thinking about your regional and long-distance journeys, what modes of transport did you use?" Previous Eurobarometer surveys do not contain the same question, which is why I use this recent survey to split the commuting flows for 2010-2018 by mode.

[20] Data source: TEN-T Regulation 2013 (Official TEN-T maps according to EU Regulation 1315/2013); TEN-T Regulation 2024 (Official TEN-T maps according to EU Regulation 2024/1679)

node closest to its geographic centroid. The shapefiles contain segment-level characteristics. For rail, they include suitability for freight or passenger transport, and whether the track is conventional or high-speed. For road, they include the road category (motorway, rural road with separate directions, rural two-lane road, urban road). I use this information to assign segment-specific travel speeds, and, for rail, to compute separate freight and passenger routes. All routes are calculated based on Dijkstra's algorithm, minimising travel time along the respective network.

The shapefiles contain information on the status of the segments, which I use to identify which segments along each optimal route were subject to project completions under the TEN-T Regulations. Specifically, the 2013 shapefile flags segments that were planned for new construction or upgrades under the 2013 TEN-T Regulation, which allows me to identify, for each optimal route, whether it traverses such a segment. Comparing with the 2024 shapefile, I determine which of these segments were subsequently completed, and for each I research the exact year of completion, drawing on a range of sources, including national government and transport authority publications, project progress reports, and media coverage. I use this information to construct a bilateral, time-varying dummy variable, separately for road and rail, taking a value of 1 if the optimal route between a given region pair traverses a segment that has been completed under a TEN-T project, from the year of completion onwards.

5.4 Calibration

Conducting counterfactual analyses using my model of regional trade and commuting formulated in changes requires data on regional bilateral trade and commuting flows by transport mode, alongside changes in trade and commuting costs implied by the infrastructure policy, as well as model parameters $\{\sigma, \rho, \eta, \theta\}$. In line with Allen and Arkolakis (2022), who set the trade elasticity to 8 for trade within the United States, also corresponding with the preferred estimate of 8.28 in Eaton and

Kortum (2002), I set the elasticity of substitution in consumption between goods from different regions to $\sigma = 9$. Fuchs and Wong (2026) estimate an elasticity of modal substitution of 1.1 between rail and truck freight within the United States, defined as the elasticity of relative modal traffic with respect to relative modal costs. In my model, this elasticity corresponds to $\rho - 1$, therefore I set $\rho = 2.1$. The estimated elasticity is substantially lower than the elasticity of trade with respect to trade costs, so substitution across modes is less elastic than substitution across origins. I set the commuting mode choice elasticity to $\eta = 3.2$ and the location choice elasticity to $\theta = 7.2$, following Herzog (2024), who estimates the elasticity of substitution between commuting by car and by public transport in London along with the elasticity of residence-workplace choice in the London area. The location choice elasticity is close to the estimate of Ahlfeldt, Redding, Sturm, and Wolf (2015) and larger than the estimates of Tsivanidis (2026) and Severen (2023).

There are two possibilities of calibrating the baseline regional trade and commuting matrix for use in my counterfactual analysis, see Dingel and Tintelnot (2026). The first one is to use the observed trade and commuting flows as baseline matrices (the "calibrated shares method"). The second one is to use the gravity estimations to create fitted baseline values of trade and commuting (the "covariates-based approach"). With the second approach requiring gravity estimations for both trade and commuting, it is not feasible due to missing mode-specific regional commuting data suitable for estimation. Therefore, I use the calibrated shares method.

The general equilibrium analysis requires a complete matrix of baseline bilateral trade and commuting flows by road and by rail, where I use averages over the pre-2013-TEN-T-regulation years 2010-2013. Domestic trade flows via road are available for all regions in the data. Domestic trade flows via rail are partly unavailable in the trade of goods dataset, which are imputed using the rail-to-road ratio where both modes are available in a given year, to obtain a complete matrix diagonal. For some regions with missing commuting data, I proxy for domestic (intraregional)

commuting using population data from the ARDECO database, which is not a perfect but a more suitable proxy than replacing missings on the matrix diagonal with zero.

6 Estimation

In this section, I estimate the changes in transport costs for rail and road induced by infrastructure projects completed under the 2013 TEN-T Regulation. Since the available data on commuting flows by transport mode is not suitable for estimation, I focus on trade flows by transport mode. I estimate the trade gravity equation from the model using the data on regional bilateral trade flows and the data on infrastructure project completions along the rail and road routes, described above.

6.1 Analysis of Variance

Before estimating the trade gravity equation, I use an analysis of variance (ANOVA) to decompose the variation in the trade data across its different dimensions. This shows how much of the variation in the dependent variable is captured by each set of fixed effects in the regression. The data on trade between EU regions by transport mode for the years 2010-2018 vary along four main dimensions: exporting region (r), importing region (k), year (t), and transport mode (l). ANOVA decomposes the variation in trade flows into shares accounted for by different factors, which can be either specific regressors of interest, such as the transport infrastructure variables used below, or sets of fixed effects. Since standard ANOVA becomes computationally demanding with high-dimensional fixed effects, I use the ANOVA-HDFE method of Adam, Larch, and Nower (2025).

I compute sequential sums of squares, which give the share of variation in regional trade flows explained by each variable as it is added sequentially to a linear regression. Because this decomposition is based on the linear regression model, I

take the natural logarithm of bilateral trade flows by transport mode, which also log-linearises the gravity equation.[21] With sequential sums of squares, the ordering of variables matters. Some orderings are dictated by the structure of the fixed effects themselves. For instance, a time-varying pair fixed effect already captures the variation of the corresponding time-invariant pair fixed effect, so the latter must be included first if both time-invariant and time-varying pair-level variation are of interest. For other dimensions, several orderings are possible. With three main sources of variation (regions, years, and transport modes) and their higher-order interactions (for example, region-year or region-pair-mode), different orderings can be considered. I start with the unilateral, region-specific variation.[22]

The ANOVAs in Figure 3 show that a third of the variation is region-specific, that is, explained by exporter and importer fixed effects. In Panel (a) of Figure 3 I next add the remaining uninteracted sources of variation, where the year dimension hardly adds any explanatory power to the regional dimension (year fixed effects explain just about 0.02% of the overall variation), while the mode fixed effects explain 45% of the remaining variation. Interacting the region and year dimension adds 0.7 percentage points to the explanatory power, while the region-mode interacted dimension matters more, adding 7.6 percentage points. The three-dimensional region-mode-year fixed effects add 0.4 percentage points to the explanatory power. The remaining variation is bilateral- and bilateral-interaction-specific, among which more than half is explained by symmetric pair fixed effects. Asymmetric pair fixed effects add 2.1 percentage points, while the next levels of pair-year and pair-mode interactions help in explaining nearly all the remaining variation, leaving only little variation at the pair-mode-year dimension.

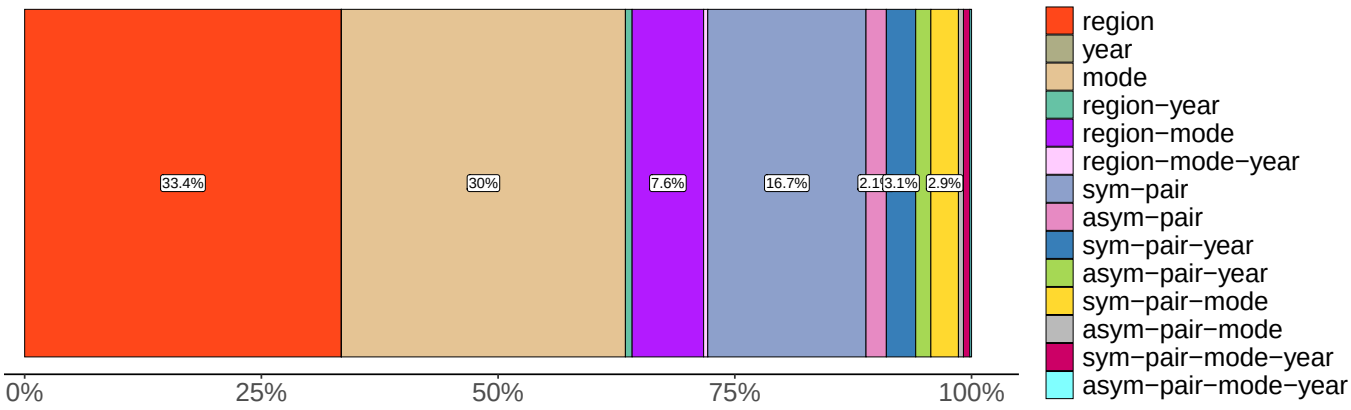
Panel (b) of Figure 3 takes a different ordering, first accounting for the unilateral(-

[21] The gravity equation could then, in principle, be estimated by OLS. Taking logarithms means that the ANOVA necessarily excludes zero trade flows, unlike the PPML regressions estimated below, which include zeros and use the recommended estimation method in the presence of zero trade flows and heteroskedasticity (Santos Silva and Tenreyro, 2006).

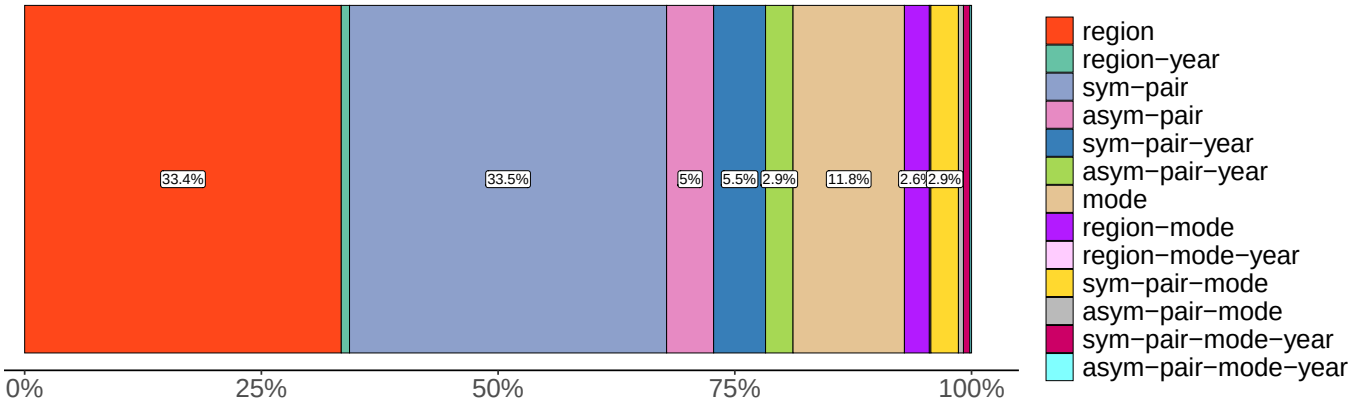
[22] I subsume exporter and importer fixed effects as region-specific variation, since there is no natural ordering among them, while their relative ordering can technically influence the results.

year) and pair(-year) dimensions, after which the mode dimension still accounts for a large part of the variation (11.8% of the total variation). Overall, the mode variation in regional trade flows matters, while a prominent role in explaining the variation is also taken by the simple region and symmetric pair dimensions. The mode and pair variation is partly overlapping, influencing the amount of variation remaining to be explained by the pair fixed effects after the inclusion of mode fixed effects and vice versa.

Figure 3: ANOVA of trade flows, fixed effects capture all dimensions of variation.



(a) Share explained in the total sum of squares. Ordering: main dimensions first, followed by interactions.



(b) Share explained in the total sum of squares. Ordering: unilateral(-time) dimensions first, followed by bilateral(-time) dimensions, mode, and interactions.

Notes: Natural logarithm of trade flows. Percentage shares of less than 2 are not labelled in the plot. Percentage shares of less than 0.0001 are not shown in the plot.

6.2 Estimating Equations

The gravity equation of trade by transport mode from Equation (17) can be transformed into the following equation of trade flows from region r to region k via mode l in year t:

$$X^l_{rk,t} = \exp\Bigg((1-\rho)\ln\left(t^l_{rk,t}\right) - (\sigma-\rho)\ln\left(\left[\sum_{l'\in\mathcal{M}}\left(t^{l'}_{rk,t}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}\right) + \ln\left(Y_{r,t}\right) + \ln\left(E_{k,t}\right) - \ln\left(Y_t\right) - (1-\sigma)\ln\left(\Pi_{r,t}\right) - (1-\sigma)\ln\left(P_{k,t}\right)\Bigg). \tag{35}$$

In order to estimate the above equation, output, expenditure, and the multilateral resistance terms are controlled for with exporter-year fixed effects, $\psi_{r,t}$, and importer-year fixed effects, $\zeta_{k,t}$. Additionally, measures for the mode-specific trade costs $t^l_{rk,t}$ and for the transport cost index $t_{rk,t} = \left[\sum_{l'\in\mathcal{M}}\left(t^{l'}_{rk,t}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}$ are needed. The aim of this study is to analyse transport infrastructure project completions for different transport modes, $TENT^{l=rail}_{rk,t}$ and $TENT^{l=road}_{rk,t}$, which are time-varying trade cost components of $t^l_{rk,t}$. The variables are assigned a value of 1 if the optimal route between regions r and k for the respective mode covers a segment that has been completed (after new construction or upgrade according to TEN-T Regulation 2013) in or before year t. They are assigned a value of 0 in any other case, that is, if the observation is not for trade via the respective mode or if the route for the respective mode has not been affected by a project completion. Time-invariant mode-specific trade cost components in $t^l_{rk,t}$ are controlled for with bilateral-pair-mode fixed effects ν^l_{rk}. The transport mode adds a dimension of variation to the bilateral-pair-year (rk,t) variation. This allows adding bilateral-pair-year fixed effects, $\chi_{rk,t}$, which control for the transport cost index $t_{rk,t}$, while not varying over transport modes (l).[23] Introducing an error term $\varepsilon^l_{rk,t}$, the estimating equation is

[23] Note that the bilateral-pair-year fixed effects can consider pairs as either symmetric (same fixed effect for flows between the pair in both directions in a given year) or asymmetric (directional).

given by

$$X^{l}_{rk,t} = \exp\left(\gamma^{rail} TENT^{l=rail}_{rk,t} + \gamma^{road} TENT^{l=road}_{rk,t} + \psi_{r,t} + \zeta_{k,t} + \nu^{l}_{rk} + \chi_{rk,t} + \varepsilon^{l}_{rk,t}\right), \tag{36}$$

where γ^{rail} and γ^{road} are the coefficients of interest to be estimated. In this theory-consistent estimating equation, γ^{rail} and γ^{road} are identified using the part of the variation in trade flows that is explained neither by the time-invariant, mode-specific dimension of trade between a given region pair (absorbed by ν^{l}_{rk}), nor by the variation common to both modes for the region pair in a given year (absorbed by $\chi_{rk,t}$). Due to the included sets of high-dimensional fixed effects, this is a demanding specification, which is why I estimate different, not entirely theory-consistent, specifications as a robustness check, separating the estimation by mode to use only bilateral-pair-year variation.

Separate equations for each transport mode, rail and road, result in an estimation model with three dimensions of variation (r, k, t), which is common in gravity estimation (see for example Yotov, Piermartini, Monteiro, and Larch, 2016; Larch and Yotov, 2024):

$$X_{rk,t} = \exp\Bigg(\tilde{\gamma} TENT_{rk,t} + \sum_{t} \tilde{\gamma}_{reg,t} regBRDR_{rk,t} + \sum_{t} \tilde{\gamma}_{int,t} intBRDR_{rk,t} + \psi_{r,t} + \zeta_{k,t} + \delta_{rk} + \tilde{\varepsilon}_{rk,t}\Bigg), \tag{37}$$

where $X_{rk,t}$ is trade via rail or trade via road (depending on the specification) from region r to region k in year t. $TENT_{rk,t}$ is a dummy variable taking a value of 1 if the optimal route between regions r and k for the mode of the respective specification covers a segment that has been completed in or before year t. $\tilde{\gamma}$ is the

In the asymmetric case, they are perfectly collinear with the exporter-year and importer-year fixed effects. By contrast, in the symmetric case, they do not fully capture the exporter-year and importer-year dimension. Therefore, the estimating equation still contains $\psi_{r,t}$ and $\zeta_{k,t}$, while acknowledging that they drop out in estimations with asymmetric pair year fixed effects.

corresponding parameter of interest. $regBRDR_{rk,t}$ are time-varying regional border dummy variables for each year in the sample, which take a value of 1 for trade between different regions within the same country for the respective year, and 0 for internal or international trade. $\tilde{\gamma}_{reg,t}$ is a corresponding parameter. $intBRDR_{rk,t}$ are time-varying international border dummy variables for each year in the sample, which take a value of 1 for trade between regions from different countries for the respective year, and 0 for internal or within-country trade. $\tilde{\gamma}_{int,t}$ is a corresponding parameter. δ_{rk} are bilateral pair fixed effects and $\tilde{\varepsilon}_{rk,t}$ is an error term.

I estimate Equations (36) and (37) using the Poisson Pseudo-Maximum-Likelihood (PPML) estimator (see Santos Silva and Tenreyro, 2006). There are different options of clustering standard errors. Typically, the gravity literature clusters standard errors in estimations with three dimensions of variation (r, k, t) by symmetric bilateral pair (see Weidner and Zylkin, 2021), while three-way clustering by exporter, importer, and year is another possibility (see Larch and Yotov, 2024). I report both types in the results for estimating Equation (37). In my estimations of Equation (36) with four dimensions of variation (r, k, l, t), one can cluster by region-pair-mode to capture correlation patters in the error terms between region pairs for a given mode over time, remaining despite the inclusion of fixed effects. Another option is four-way clustering by exporter, importer, mode, and year to allow for $cov\left(\varepsilon_{rkl,t}, \varepsilon_{rfm,d}\right) \neq 0$, $cov\left(\varepsilon_{rkl,t}, \varepsilon_{jkm,d}\right) \neq 0$, $cov\left(\varepsilon_{rkl,t}, \varepsilon_{jfl,d}\right) \neq 0$, and $cov\left(\varepsilon_{rkl,t}, \varepsilon_{jfm,t}\right) \neq 0$ (see Larch and Yotov, 2024 for a similar reasoning in panels with three dimensions of variation). A third option is inspired by Weidner and Zylkin (2021) and clusters two-way by region-pair-year and region-pair-mode.

6.3 Estimation Results

Results from estimating Equation (36) are shown in Table 1 and ANOVAs for the corresponding specifications are shown in Figure 4.[24] The ANOVAs show that the bilateral-pair-year and bilateral-pair-mode fixed effects capture different dimensions of the variation in trade flows and both are relevant, each explaining substantial shares of the variation. In the first specification, shown in Column (1) of Table 1, I use asymmetric (directional) region pairs for creating the fixed effects $\chi_{rk,t}$ and ν^{l}_{rk}, giving a coefficient estimate for a completed rail infrastructure project between two regions of about 0.18, while that for a road project is about 0.04. In Column (2) the bilateral pair fixed effects are symmetric (i.e. the same fixed effect for both directions of flows between two regions) and the coefficients are of a similar magnitude, with the rail coefficient slightly smaller and the road coefficient slightly larger than in Column (1). In both specifications, the positive effect of rail projects on bilateral regional trade flows is significant for standard errors clustered by exporter, importer, and year as well as two-way clustering, while the estimate is smaller and not significant for road projects.

To check for robustness, I estimate Equation (37), separating the trade flow data by rail and by road. The results in Table 2 are shown for three specifications for each mode. Columns (1) and (4) use asymmetric pair fixed effects, while Columns (2) and (5) use symmetric pair fixed effects. In Columns (3) and (6) I exploit the regional structure of the data that is nested into countries, adding asymmetric-country-pair-year fixed effects in addition to the symmetric pair fixed effects. The estimated coefficient for TEN-T rail projects is positive throughout specifications, becoming significant when using symmetric pair fixed effects in Columns (2) and

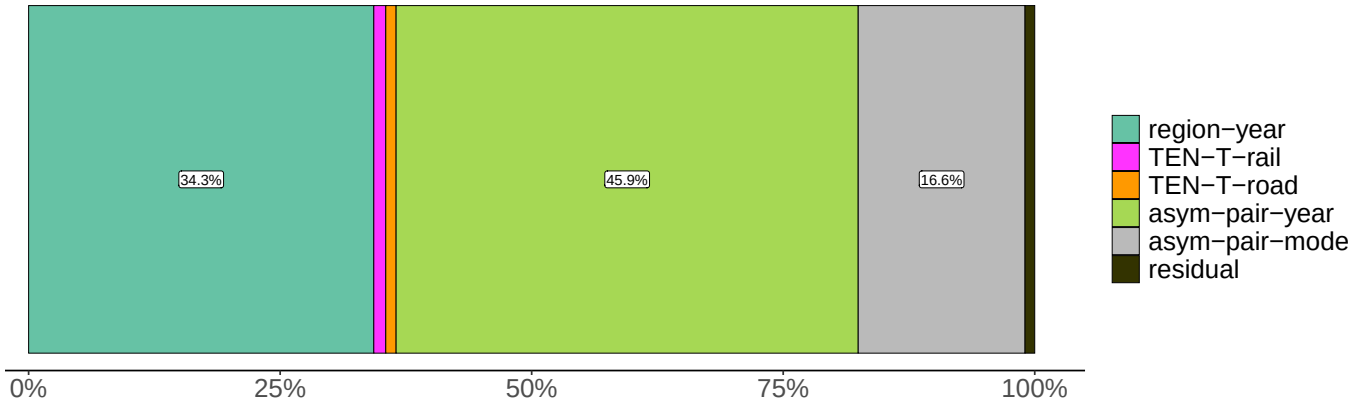
[24]Note that the samples for the PPML estimations and the ANOVAs differ, as ANOVA excludes zero trade flows. Further, ANOVAs show that the two variables of interest, $TENT^{rail}_{rk,t}$ and $TENT^{road}_{rk,t}$, exhibit variation beyond the entire set of included fixed effects. Their variation is, however, reduced below a level to which it can be visualised in the barplot if they are included last. For this reason, the specifications shown in Figure 4 include them after the exporter-year and importer-year fixed effects, to show the amount of the bilateral-mode-year variation they explain.

Table 1: PPML Estimation Results, Estimating Equation (36).

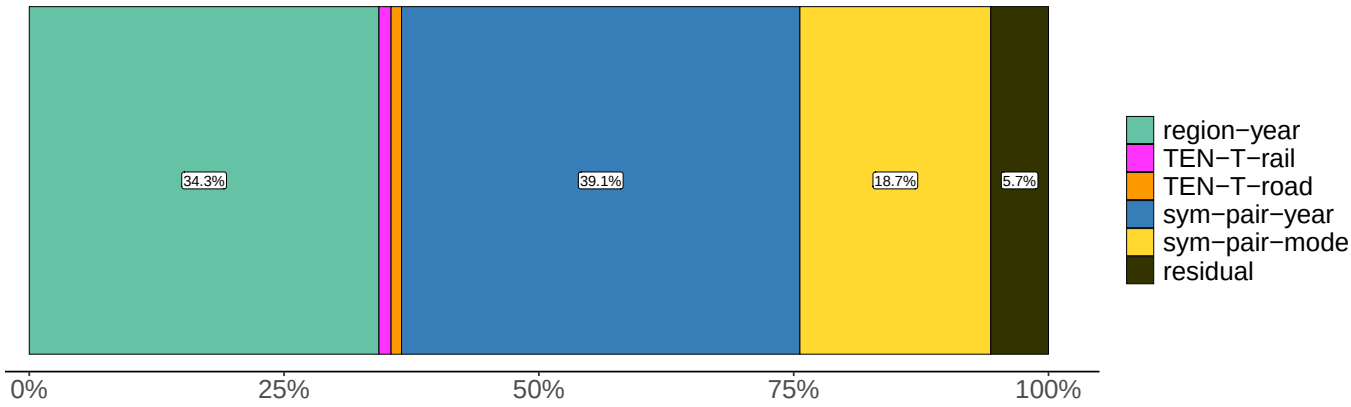
	(1)	(2)
TENT^{rail}	**0.1829**	**0.1685**
SE clu. symm. pair-mode	(0.1780)	(0.1287)
SE clu. exp/imp/mode/y	(0.0559)***	(0.0547)***
SE clu. symm. pair-year/ symm. pair-mode	(0.1077)*	(0.1063)
TENT^{road}	**0.0379**	**0.0433**
SE clu. symm. pair-mode	(0.0509)	(0.0365)
SE clu. exp/imp/mode/y	(0.0628)	(0.0652)
SE clu. symm. pair-year/ symm. pair-mode	(0.0407)	(0.0396)
Fixed-effects:		
Exporter-year	$^{(a)}$	Yes
Importer-year	$^{(a)}$	Yes
Asymmetric-pair-year	Yes	
Asymmetric-pair-mode	Yes	
Symmetric-pair-year		Yes
Symmetric-pair-mode		Yes
Observations	698,039	761,664
Pseudo R^2	0.99255	0.97450

Notes: Clustered standard-errors (SE clu.) in parentheses, with symm. pair-mode denoting clustering by symmetric-region-pair-mode, exp/imp/mode/y denoting four-way clustering by exporter & importer & mode & year, symm. pair-year/symm. pair-mode denoting two-way clustering by symmetric-region-pair-year & symmetric-region-pair-mode.
Significance codes: *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$.
$^{(a)}$: Asymmetric-pair-year fixed effects make the use of exporter-year and importer-year fixed effects redundant.

Figure 4: ANOVA of trade flows, specifications from Table 1.



(a) Share explained in the total sum of squares. Covariates and fixed effects from Column (1) in Table 1.



(b) Share explained in the total sum of squares. Covariates and fixed effects from Column (2) in Table 1.

Notes: Natural logarithm of trade flows. Percentage shares of less than 2 are not labelled in the plot. Percentage shares of less than 0.0001 are not shown in the plot.

Table 2: PPML Estimation Results, Separate Regressions for Trade via Rail and Road, Estimating Equation (37).

Dependent Variable:	Trade via rail			Trade via road		
	(1)	(2)	(3)	(4)	(5)	(6)
TENT	**0.0456**	**0.0861**	**0.0480**	**-0.0102**	**-0.0106**	**-0.0481**
SE clu. symm. pair	(0.0385)	(0.0355)**	(0.0198)**	(0.0173)	(0.0173)	(0.0177)***
SE clu. exp/imp/y	(0.0554)	(0.0231)***	(1×10^{-5})***$^{(b)}$	(0.0195)	(0.0196)	(0.0190)**
Fixed-effects:						
Exporter-year	Yes	Yes	Yes	Yes	Yes	Yes
Importer-year	Yes	Yes	Yes	Yes	Yes	Yes
Asymmetric-pair	Yes			Yes		
Symmetric-pair		Yes	Yes		Yes	Yes
Asym.-country-pair-year			Yes			Yes
Further variables:						
RegBRDR-year	Yes	Yes	Yes	Yes	Yes	Yes
IntBRDR-year	Yes	Yes	$^{(c)}$	Yes	Yes	$^{(c)}$
Observations	345,491	370,951	366,417	365,121	397,683	397,484
Pseudo R^2	0.97997	0.95924	0.96347	0.94979	0.94171	0.94319

Notes: Clustered standard-errors (SE clu.) in parentheses, with symm. pair denoting clustering by symmetric bilateral region pair, exp/imp/y denoting clustering by exporter & importer & year.
Significance codes: *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$.
$^{(b)}$: Variance contained negative values in the diagonal and was "fixed" according to Cameron, Gelbach, and Miller, 2011.
$^{(c)}$: Asymmetric-country-pair-year fixed effects control for time-varying international borders.

(3). The estimated coefficient for TEN-T road projects does not appear to be robust, switching signs to a negative estimate throughout specifications, becoming significant only in the specification with symmetric pair and asymmetric country pair year fixed effects. Overall, the results suggest that completed TEN-T rail projects have a robust positive effect on regional bilateral trade flows, while there is no clear effect for TEN-T road projects.

6.4 Accounting for Construction and Adjustment: 3-Year Averages

Transport infrastructure projects typically involve construction sites, such that certain routes may not be navigable while constructions are ongoing. This may reduce trade along the optimal route before project completion, potentially especially in the year of completion. After completion, trade routes may take a while to adjust. To account for this potential temporary reduction in trade flows, I create 3-year averages of my data. The dummy variables $TENT^{rail}_{rk,t}$ and $TENT^{road}_{rk,t}$ are assigned the value 1 if the optimal route between regions r and k for the respective mode covers a segment that has been completed in or before all years in the 3-year period t.

Table 3 reports results from estimating Equation (36) and Table 4 reports results from estimating Equation (37) with PPML on the 3-year-averaged data. Qualitatively, the previous results are confirmed. While rail project completions have a robust positive effect on regional trade flows, the effect of road project completions remains unclear, with signs switching across specifications. In magnitude, the rail project completion effect is larger in the 3-year-averaged data than in the yearly data, suggesting that the estimates for yearly data were partly capturing temporary construction-related disruptions to trade. The results from Table 3 suggest that TEN-T a rail project completion along the optimal route between two regions increases trade flows by about 27%.[25] The unclear results for road project completions can be explained by the TEN-T road network already being comparatively mature at the outset of the sample period, such that a completed project brings only a small marginal improvement. The rail network, by contrast, faced more substantial improvements still to be made, along with eight times larger funding for rail than for road projects under the 2013 Regulation.

Overall, estimating the theory-consistent gravity equation (36) and a separate

[25] $[\exp(0.24) - 1] \times 100 = 27.12$

gravity equation (37) for rail shows that TEN-T rail project completions have a positive effect on bilateral regional trade flows. While there are differences in magnitude for the estimated coefficients across both estimating equations, the positive effect is robust. Note that the equations are not estimating exactly the same effect, due to the different samples and fixed effects used, so a direct comparison is not suitable. To recover the structural estimate of a rail transport cost reduction due to TEN-T project completions, I will therefore use the results from the theory-consistent estimation on 3-year-averaged data.

6.5 Trade and Commuting Cost Changes due to TEN-T Policy

The above estimations suggest that TEN-T rail projects have a robust positive effect on regional trade flows, while road projects do not show a clear effect. Hence, I base my counterfactual TEN-T policy analysis on the rail projects, consistent with rail's status as a TEN-T policy priority. In this section, I recover the changes in rail transport costs due to completed TEN-T rail projects, based on my theory-consistent estimation results.

Equation (35) shows that the rail-specific transport costs are not only contained in the term $(1-\rho)\ln\left(t^{l}_{rk,t}\right)$ but also in the term $-(\sigma-\rho)\ln\left(\left[\sum_{l'\in\mathcal{M}}\left(t^{l'}_{rk,t}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}\right)$. Since the latter is controlled for by the fixed effects when estimating Equation (36), I approximate the change in rail-specific trade costs due to TEN-T rail project completions using the following:

$$\begin{aligned}
(1-\rho)\ln\left(\widehat{t}^{rail}_{rk,t}\right) &= \gamma^{rail}TENT^{rail}_{rk,t} \\
\Leftrightarrow \qquad \left(\left(\widehat{t}^{rail}_{rk,t}\right)^{(1-\rho)}\right) &= \exp\left(\gamma^{rail}TENT^{rail}_{rk,t}\right) \\
\Leftrightarrow \qquad \widehat{t}^{rail}_{rk,t} &= \exp\left(\gamma^{rail}TENT^{rail}_{rk,t}\right)^{\frac{1}{1-\rho}}
\end{aligned}$$

Using the coefficient estimate from Column (1) in Table 3 based on the 3-year

Table 3: 3-Year-Averaged Data, PPML Estimation Results, Estimating Equation (36).

	(1)	(2)
TENT^{rail}	**0.2403**	**0.2470**
SE clu. symm. pair-mode	(0.1918)	(0.1257)**
SE clu. symm. pair-year/ symm. pair-mode	(0.1139)**	(0.1066)**
TENT^{road}	**0.0947**	**-0.1771**
SE clu. symm. pair-mode	(0.0474)**	(0.0639)***
SE clu. symm. pair-year/ symm. pair-mode	(0.0318)***	(0.0564)***
Fixed-effects:		
Exporter-period	[a]	Yes
Importer-period	[a]	Yes
Asymmetric-pair-period	Yes	
Asymmetric-pair-mode	Yes	
Symmetric-pair-period		Yes
Symmetric-pair-mode		Yes
Observations	252,663	276,325
Pseudo R^2	0.99263	0.97674

Notes: "Period" refers to the three-year period over which the data was averaged, that is, one of 2010-2012, 2013-2015, 2016-2018.
Clustered standard-errors (SE clu.) in parentheses, with symm. pair-mode denoting clustering by symmetric-region-pair-mode, symm. pair-year/symm. pair-mode denoting two-way clustering by symmetric-region-pair-year & symmetric-region-pair-mode.
With 3-year averaged data over the data coverage of 2010-2018, the number of clusters by period is just 3. As cluster-robust inference is typically suitable for larger numbers of clusters (see Cameron, Gelbach, and Miller, 2011), standard errors clustered by exporter & importer & mode & period are not reported.
Significance codes: *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$.
[a]: Asymmetric-pair-period fixed effects make the use of exporter-period and importer-period fixed effects redundant.

Table 4: 3-Year-Averaged Data, PPML Estimation Results, Separate Regressions for Trade via Rail and Road, Estimating Equation (37).

Dependent Variable:	Trade via rail			Trade via road		
	(1)	(2)	(3)	(4)	(5)	(6)
TENT	**0.0505**	**0.1206**	**0.0784**	**-0.0430**	**-0.0903**	**-0.1077**
SE clu. symm. pair	(0.0406)	(0.0378)***	(0.0221)***	(0.0199)**	(0.0197)***	(0.0205)***
Fixed-effects:						
Exporter-period	Yes	Yes	Yes	Yes	Yes	Yes
Importer-period	Yes	Yes	Yes	Yes	Yes	Yes
Asymmetric-pair	Yes			Yes		
Symmetric-pair		Yes	Yes		Yes	Yes
Asym.-country-pair-period			Yes			Yes
Further variables:						
RegBRDR-period	Yes	Yes	Yes	Yes	Yes	Yes
IntBRDR-period	Yes	Yes	(c)	Yes	Yes	(c)
Observations	117,526	128,492	128,041	137,189	148,683	148,683
Pseudo R^2	0.98081	0.95990	0.96320	0.97133	0.96067	0.96204

Notes: "Period" refers to the three-year period over which the data was averaged, that is, one of 2010-2012, 2013-2015, 2016-2018.
Clustered standard-errors (SE clu.) in parentheses, with symm. pair denoting clustering by symmetric bilateral region pair.
With 3-year averaged data over the data coverage of 2010-2018, the number of clusters by period is just 3. As cluster-robust inference is typically suitable for larger numbers of clusters (see Cameron, Gelbach, and Miller, 2011), standard errors clustered by exporter & importer & period are not reported.
Significance codes: *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$.
(c): Asymmetric-country-pair-period fixed effects control for time-varying international borders.

averaged data, $\tilde{\gamma}^{rail} = 0.24$, together with $\rho = 2.1$ (see Section 5.4), this gives $\widehat{t}_{rk,t}^{rail} = \exp\left(\tilde{\gamma}^{rail} TENT_{rk,t}^{rail}\right)^{\frac{1}{1-\rho}} = \exp\left(0.24\right)^{\frac{1}{1-2.1}} \approx 0.80$ for the change in rail transport costs between two regions when a rail infrastructure project is completed $\left(TENT_{rk,t}^{rail} = 1\right)$. In my counterfactual analysis, this reduction is assigned to the trade costs for trade via rail between two regions whose optimal shipping route was affected by a completed freight rail project from the 2013 TEN-T Regulation, as in the data.

Since no data is available at the regional level on the use of transport modes for regional bilateral commutes, I use country-level shares from survey data for obtaining baseline commuting flows (see Section 5). While they can serve as a proxy for baseline commuting shares, the obtained flows are not suitable for regressions of the effect of rail and road transport infrastructure project completions on bilateral commuting flows. I therefore assume that the changes in the costs of transportation for trade via rail also apply to the changes in rail commuting costs. This approach can be seen as conservative, since trade flows are observed between regions throughout Europe and trade via rail is typically used for longer distances, while commuting in my setting is only possible within the same country or neighbouring countries. Hence, the share in saved travel time through an infrastructure project completion for a typical rail commute is plausibly larger than that saved for a longer typical rail freight. In my counterfactual analysis, I thus assign a change of $\widehat{d}_{kn,t}^{rail} = 0.80$ to the commuting costs via rail between two regions whose optimal commuting route was affected by a completed passenger rail project from the 2013 TEN-T Regulation, as in the data.[26]

[26] Using the same cost changes for trade and commuting also allows straightforward comparison of the relative contribution of the trade versus commuting cost reduction to the aggregate welfare effects. In this case, differences in the contribution to the aggregate welfare effect are not attributable to different strengths of the shock. In future work, different cost reductions for trade and commuting can be analysed.

7 General Equilibrium Analysis

As the system of Equations (22)-(29) is homogeneous of degree zero in prices, I choose a numéraire region, the Greek region EL54 (Epirus). The region is not directly affected by any completed rail project, for two reasons. On the one hand, it is located on the Eastern outskirts of the TEN-T network and does not contain any new constructions or upgrades. On the other hand, when an optimal route between Epirus and another regions traverses an affected segment, trade or commuting flows via rail are zero, such that the change does not effectuate in the implementation in changes, as this captures only the intensive margin and not the extensive margin. All nominal effects from the general equilibrium analysis are to be interpreted relative to the Greek region. For this reason, most of the results shown in the following section are real variables, except for trade flow changes, while the results for nominal variables can be found in Appendix C.

7.1 General Equilibrium Effects of the 2013 TEN-T Regulation's Rail Projects

Consistent with TEN-T's stated priority of shifting freight and passenger transport onto rail and with the estimation results, I quantify the general equilibrium effects of rail transport cost reduction from completed TEN-T rail projects.

I start by reporting effects on trade and commuting flows, as they are immediately affected by the transport cost reduction. For trade, comparing the change in nominal total regional road exports in Panel (a) of Figure 5 with the change of rail exports in Panel (b) of Figure 5 shows that road exports tend to decrease by up to 8.71% for some regions, while rail exports increase for most regions and do so very strongly for some regions, especially in Italy, Spain, and Germany, by more than 100%.[27] On average, regional road exports increase by about 0.70% while regional

[27]Nominal changes depend on the numéraire, the Greek region Epirus. I nonetheless report nominal change in trade flows, as they are informative about the relative adjustment of trade flows

rail exports increase by about 7.52%. Although shipping via road tends to be substituted by shipping via rail after the rail transport cost reduction, road exports still increase on average. This means that the lower trade cost index (which accounts for the reduced rail transport costs) increases overall trade by more than the substitution to rail decreases the road shipping. The changes for aggregate exports and imports, as well as for imports via road and rail, are shown in Appendix Figures C.1 and C.2. Domestic sales tend to slightly decrease for many regions, while some regions encounter relatively strong increases, such that on average, regional domestic sales increase by 0.46%, as shown in Appendix Figure C.3.

For commuting, I compare the change in total outcommuters from all regions for the different modes, road and rail, shown in Panels (a) and (b) of Figure 6. Strong effects are particularly observed in those countries where rail infrastructure completions affect within-country commuting costs (Germany, France, Italy, Spain, Hungary, Denmark). While the outwards commutes from regions tend to increase on average for both, road (3.23%) and rail (14.85%), there are also some regions for which both slightly decrease, for example for all Swedish regions. Changes for incommuting via road and rail exhibit similar patterns and are shown in Appendix Figure C.4. Changes for commuting flows aggregated over transport modes are shown in Appendix Figure C.5. More people tend to move (either living and working in different regions or migrating), shown in the decrease in the number of domestic workers in Appendix Figure C.6 in most regions, while some regions encounter strong increases.

Commuting flows are the result of workers' choices over where to live and where to work. The regional labour force changes are shown in Panel (a) of Figure 7. It shows decreases of up to 0.97% for many regions, while some regions encounter relatively strong labour force increases of up to 9.74%, with an average decrease of 0.08%. A very similar pattern is observed in the changes of regional residents in

after the transport cost reduction.

Panel (b) of Figure 7. This similar pattern *in general* is explained by many people living and working in the same region, while the fact that they both exhibit similar *changes* can be explained by migration, which is implicit in the framework and given by workers who choose to reside in a different region in the counterfactual than in the baseline. Setting both into relation, Figure 8 shows the change in the ratio of workers per resident. While many regions encounter only minor changes (pointing at workers who live and work in the same region relocating to another region to live and work) and the average change is a decrease of 0.02%, some regions, especially in countries affected by commuting cost changes (Germany, France, Italy, Spain, Hungary, Denmark) exhibit increases of up to 6.78% and decreases of more than 7%.

In the framework with commuting, workers earn their wage in their region of work, while they spend it on consumption goods in their region of residence, which can be different regions. Dividing regional wages by regional prices gives real wages only for the group of workers living and working in the same region, while neglecting the commuters who live and work in different regions. The average real wage change for each workplace, weighting the regional real wage changes by commuting shares for all residence-workplace pairs, is shown in Panel (a) of Figure 9. Real wages increase on average by 0.05%, while some regions (especially in Hungary, Spain, and Italy) exhibit increases of up to 1.22% and some regions encounter small decreases of up to 0.29% (especially in Germany, Denmark, and France). The average change of real GDP for each workplace tends to be negative for many regions, while some regions encounter relatively strong increases, giving an average moderate decrease of 0.03%, as shown in Panel (b) of Figure 9. Note that the pattern of labour force changes observed in Panel (a) of Figure 7 is nearly identical to that of real GDP changes (albeit on a different scale). The contrast between observed average real wage increases and real GDP decreases can be explained by the decreasing labour force in many regions, recalling that labour is the only factor of production in the

framework.

Nominal wages and consumer prices increase in nearly all regions, shown in Appendix Figure C.7. Changes of nominal output and nominal expenditure are shown in Appendix Figure C.8.

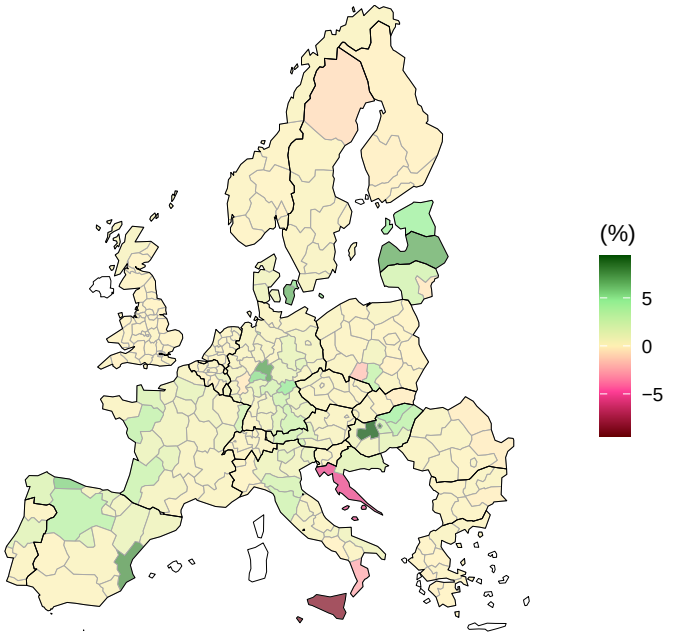
Overall, the patterns for trade and commuting flows are as expected, noting that rail use is increased substantially, while road use partly encounters small increases due to reduced transport cost indices and general equilibrium effects. There is substantial heterogeneity in the effects, showing that the rail infrastructure policy affects regions differently, such that heterogeneity is not only observed across but also within countries.

7.2 Welfare Effects

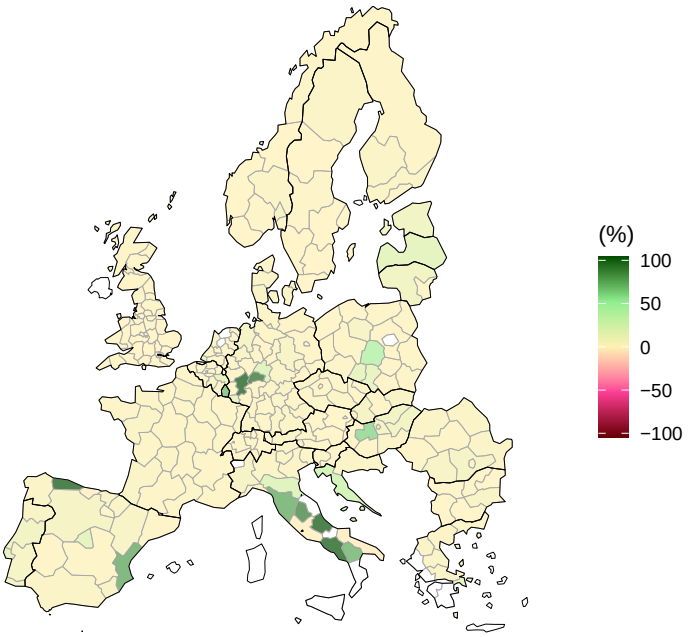
Transport infrastructure projects are costly, raising the question of whether their aggregate welfare gains are large enough to justify the investment. This section quantifies the aggregate welfare gains from the rail infrastructure improvements under the 2013 TEN-T policy, using the results from the GE analysis which isolate the counterfactual effects if nothing else apart from the rail transport cost reduction had happened. In a back-of-the-envelope assessment, I convert the welfare gains into a monetary value using equivalent variation and relate it to the rail infrastructure investment. I also disentangle the contribution of the trade versus commuting cost reduction in explaining the overall effects.

As a measure of welfare I calculate the change in ex ante expected utility as in Equation (34), following recent studies, such as Allen and Arkolakis (2022), Herzog (2024), Dingel and Tintelnot (2026), and Tsivanidis (2026). Expected utility is homogeneous of degree one in real wages, so the change in expected utility corresponds with the equivalent variation as a share of baseline income. It is the uniform proportional increase in real income, which would make workers as well off as the counterfactual rail transport cost reduction.

Figure 5: GE Results of 2013 TEN-T rail projects: Change in exports via road and rail.



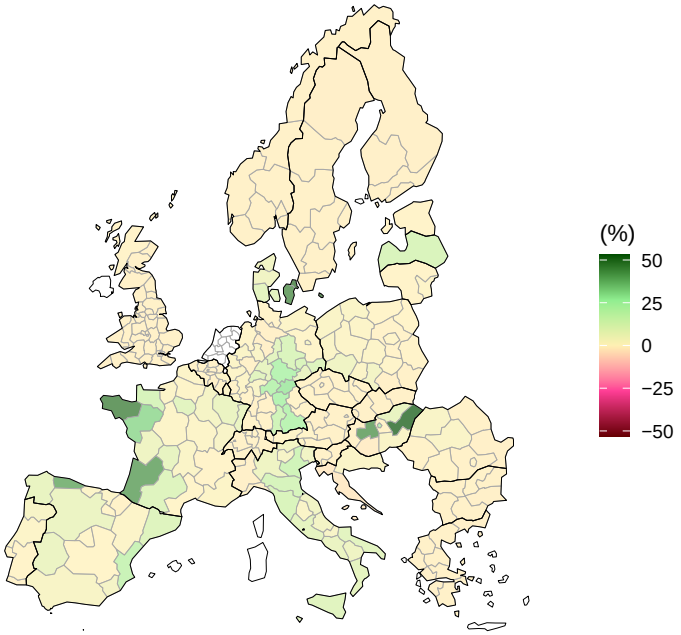
(a) Road exports change.



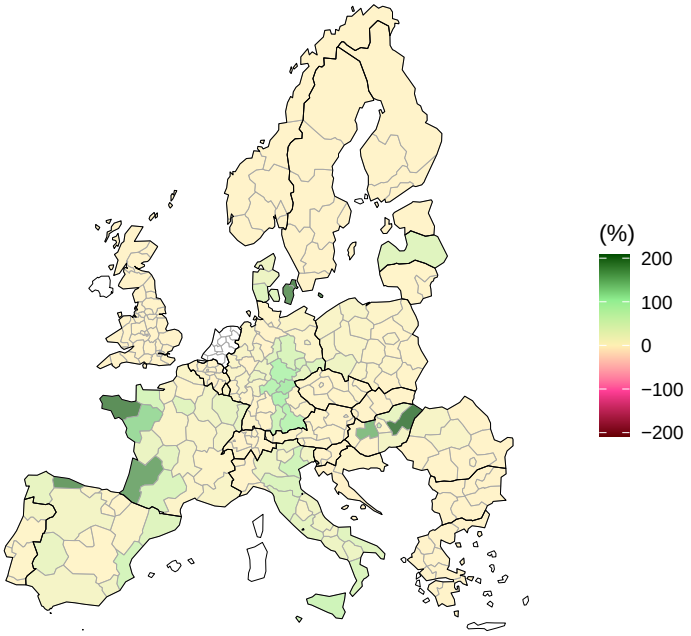
(b) Rail exports change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.
Panel (b): For the following regions, the change in rail exports is not coloured to scale: Spanish region Asturias (ES12, 129.25%), German region Koblenz (DEB1, 153.93%), Italian region Campania (ITF3, 400.85%).

Figure 6: GE Results of 2013 TEN-T rail projects: Change in outcommuters via road and rail.



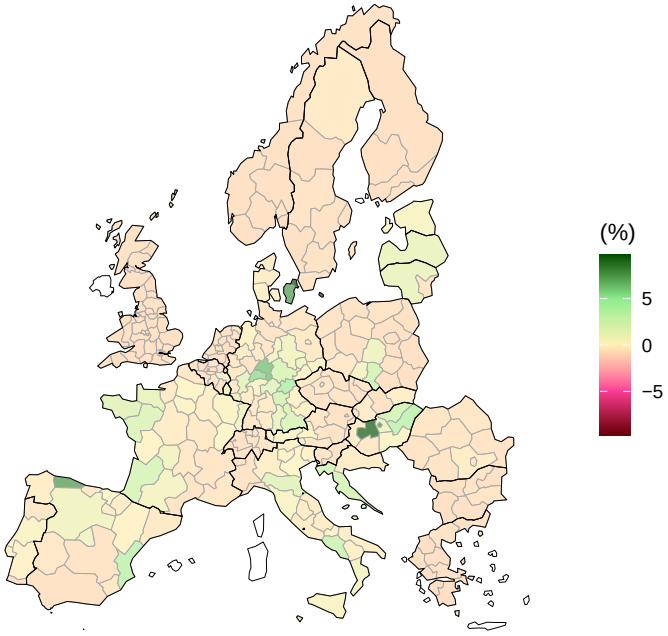
(a) Outcommuters via road change.



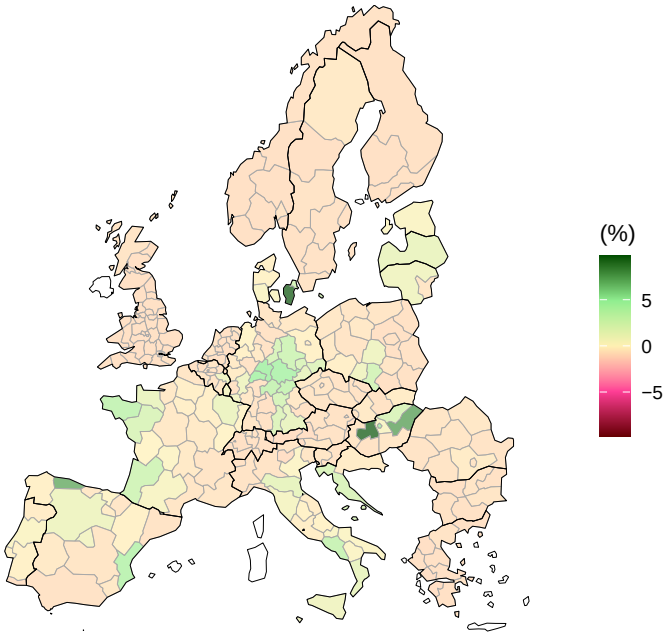
(b) Outcommuters via rail change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure 7: GE Results of 2013 TEN-T rail projects: Change in labour force and residents.



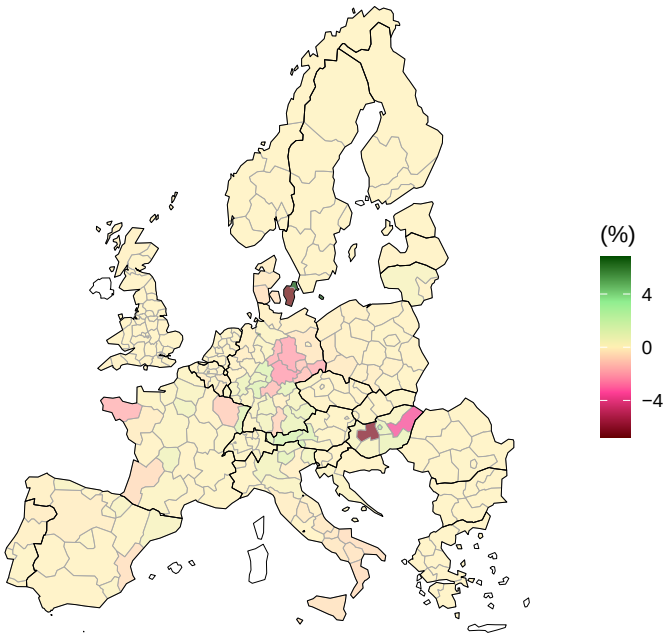
(a) Labour force change.



(b) Residents change.

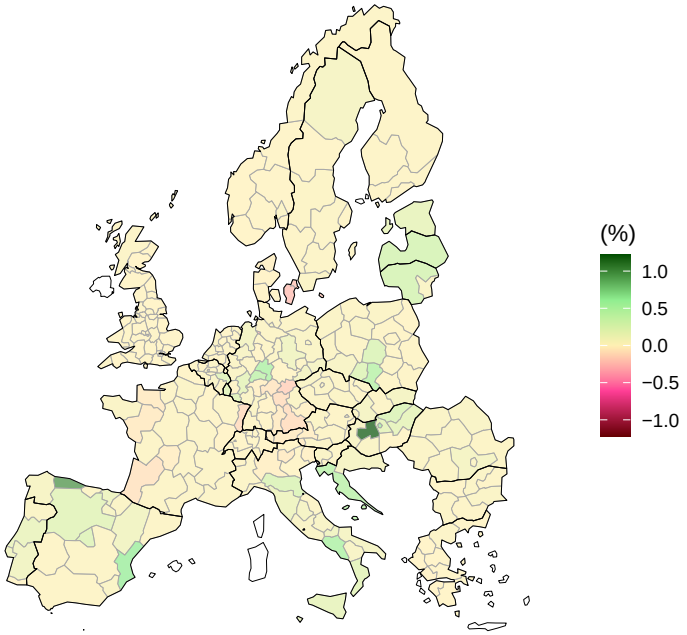
Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.
Panel (b): For the following regions, the change in residents is not coloured to scale: Hungarian region Közép-Dunántúl (HU21, 16.92%), Danish region Zealand (DK02, 21.61%).

Figure 8: GE Results of 2013 TEN-T rail projects: Change in workers per resident.

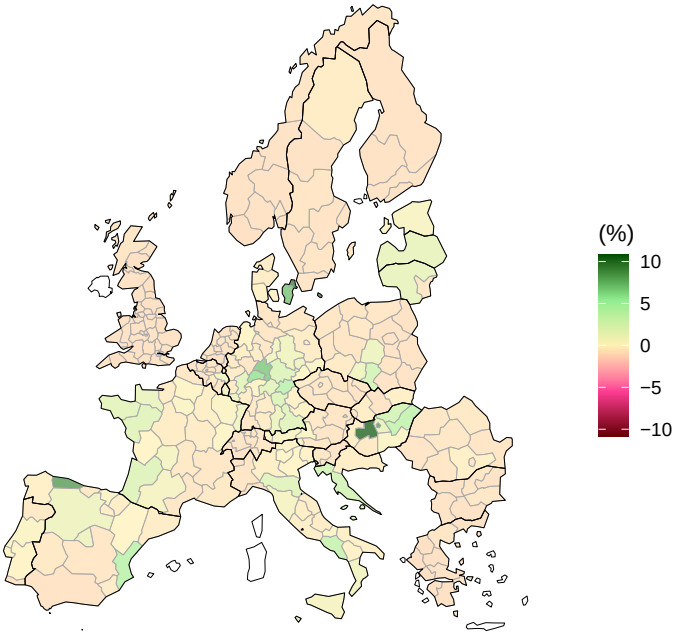


Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data. For the Danish region Zealand (DK02), the change in workers per resident of -11.67% is not coloured to scale.

Figure 9: GE Results of 2013 TEN-T rail projects: Change in real wages and real GDP (commuting-weighted average by workplace).



(a) Real wages change.



(b) Real GDP change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Using my GE results, I obtain a welfare increase of 0.13%. Interpreting the counterfactual equilibrium as the annual outcome with the reduced rail transport costs from completed TEN-T projects, I apply the welfare increase to annual output. Multiplying with baseline output in my framework, I obtain a back-of-the-envelope annual monetary benefit of about 9.1 billion euros.[28] To assess the size of this benefit, I compare it with the rail investment for the realised TEN-T rail projects. The EU allocated 16.24 billion euros to the rail infrastructure projects under the 2013 Regulation, which acts as a catalyst, while Member States provide most of the funding. Applying a leverage factor of about 2–3 (see Marketa, 2021) to the EU's contribution of 16.24 billion euros, I estimate total rail investment at 32–49 billion euros. Abstracting from discounting, the yearly benefit of 9.1 billion euros covers the investment in roughly 4–5 years. This simplified back-of-the-envelope calculation gives an impression for the benefits captured by my framework, which materialise only through a reduction in rail freight and rail commuting costs, and only cover goods trade within and across the regions in the sample. Other aspects of the economy (e.g. services) or non-monetary effects (e.g. environmental externalities), as well as temporal discounting, may be considered in a full cost-benefit analysis.

Performing additional counterfactual analyses in which only trade costs are reduced, while holding commuting costs unchanged, or in which only commuting costs are reduced, while holding trade costs unchanged, I can compare the importance of the transport infrastructure improvement for trade and for commuting in explaining the overall welfare effects. Calculating the change in expected utility, I obtain a welfare increase of 0.05% in the scenario where only trade costs are reduced and of 0.08% in the scenario where only commuting costs are reduced. Remarkably, the

[28]Baseline output in my implementation is 6.9 trillion euros and consists of the total value of goods trade within and across the 271 NUTS2 regions in the sample, averaged over 2010–2013, which is substantially lower than GDP for the EU27 reported for each of the years 2010–2013, amounting to over 11 trillion euros (see https://ec.europa.eu/eurostat/databrowser/view/nama_10_gdp__custom_22840297/default/table). Assuming that the effects apply more broadly to the entire EU and further activities (e.g. services), yearly monetary benefits could be substantially higher.

reduction in commuting costs explains more than half, about 62%, of the overall welfare effect of 0.13% for the scenario combining trade and commuting cost reductions, leaving about 38% to be explained by the reduction in trade costs.[29]

Although there is no comparable study quantifying the general equilibrium welfare effects for TEN-T rail projects for trade and commuting, I present some findings from the literature for context. Focusing on trade and the regional effects of EU Cohesion Policy road infrastructure investment, Shevtsova, Díaz-Lanchas, Persyn, and Mandras (2025) find an EU-wide average welfare increase of 0.037%. The welfare gain of 0.05% for an isolated rail trade cost reduction I obtain is larger, staying roughly in the same order of magnitude in the European context. Focusing on the commuting cost reduction only, the welfare increase of 0.08% lies in the same order of magnitude as the 0.11% welfare increase Herzog (2024) obtains for the introduction of London's Congestion Charge in a framework with commuting, mode choice, and traffic. A scenario with a similar commuting cost reduction as in my counterfactual analysis in Monte, Redding, and Rossi-Hansberg (2018), implemented as a common 21% commuting cost reduction for all US counties, yields a welfare gain of 6.89%. This effect is much stronger than the welfare effect I obtain for an isolated rail commuting cost reduction, which is not surprising, considering Monte, Redding, and Rossi-Hansberg (2018) do not target an individual transport mode and implement the commuting cost reduction between all US counties. By contrast, in my application, only 23% of all possible commutable region pairs (in the same or neighbouring countries) are affected by a commuting cost reduction only for rail. Excluding commutable region pairs with zero commuting flows, which imply that commuting cost reductions cannot materialise into an increase in commuting by construction, this share reduces to 15%. This context highlights the potential of transport infrastructure policy for commuting. If commuting costs are reduced between more region pairs and if one allows for new commuting links to emerge, the

[29] Interaction effects are negligibly small.

welfare effects may well increase.

Finally, comparing my results with a more conservative counterfactual of a reduction of 4.5% in trade costs and commuting costs ($\widehat{t}^{rail}_{rk,t} = 0.955$, $\widehat{d}^{rail}_{kn,t} = 0.955$), inspired by the lowest coefficients from the separate regressions for rail, gives a welfare increase of about 0.018%. Although this is a moderate increase, the relative role of commuting in bringing across the overall welfare change remains important, with commuting explaining more than half of the welfare effect (56%), leaving less than half (44%) to be explained by the trade cost reduction.

8 Conclusion

This paper evaluates the spatial effects and welfare implications of large-scale transport infrastructure investments in road and rail across Europe. I develop a structural framework combining goods trade and worker commuting via different transport modes, road and rail. The framework spans multiple countries and their subnational regions, with trade and commuting both able to cross national borders. This combination makes it possible to apply the framework to the ambitious Europe-wide EU TEN-T infrastructure policy, which targets cross-border links and different transport modes. Together with novel data on regional bilateral trade and commuting flows, as well as on TEN-T rail and road infrastructure projects across Europe, I quantify the general equilibrium effects of the 2013 TEN-T rail and road policy.

My structural gravity estimation results suggest that completed TEN-T rail infrastructure projects considerably reduce rail transport costs, while road projects show no robust effect. Applying the estimated rail transport cost reduction to trade and commuting in a counterfactual general equilibrium analysis shows that it increases welfare by 0.13% across subnational regions in Europe. Decomposing the effect, I find that commuting explains the larger share of the welfare effect (62%), compared to goods trade (38%). Policy evaluations of infrastructure investment

should therefore incorporate labour mobility alongside trade to avoid underestimating its economic returns. A back-of-the-envelope comparison of these welfare gains to total investment into rail projects under the 2013 TEN-T Regulation suggests that the annual benefit covers the investment in roughly 4–5 years.

The general equilibrium effects reveal substantial regional heterogeneity across and within countries, in trade flows, commuting flows, real wages, and the spatial allocation of workers between residence and workplace regions. Such regional heterogeneity remains hidden in country-level analyses, highlighting the importance of a subnational pan-European perspective when evaluating EU-wide policy such as TEN-T. The results are relevant for the EU's ongoing TEN-T policy, which aims to improve cross-border links for transporting both goods and people, the two channels my framework combines, as well as regional cohesion. My framework can also be used to evaluate the ex ante effects of projects under the revised 2024 TEN-T Regulation or appraise future projects before they are implemented.

TEN-T's further objectives for building an efficient and sustainable multimodal network suggest several extensions of my framework. Allowing for optimal routing, as in Allen and Arkolakis (2022), would let transport cost changes affect routing decisions around improved or congested segments. Adding traffic congestion would address the trade-off between capacity expansion and induced congestion that is central to infrastructure policy in practice. Allowing switches between modes at intermediate nodes, as in Fuchs and Wong (2026), would let trade and commuting use the multimodal feature of TEN-T. Finally, including emissions by transport mode could extend the evaluation of infrastructure investment beyond the economic returns, as decarbonisation is a further objective of TEN-T policy.

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Appendix

A Model Details and Derivations

A.1 Derivations for Consumer Demand

Sourcing Decision: Origin of Consumption Goods

With Fréchet-distributed preference shocks and iceberg-type commuting costs, utility from (1) can be written as

$$U_{kn}^{m}(\omega) \propto \left[\sum_{r \in \mathcal{R}} \left(\left[\sum_{l \in \mathcal{M}} \left(c_{rk}^{l} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \tag{A.1}$$

The worker chooses which regions to purchase her consumption goods from to maximise CES utility subject to the budget constraint:

$$\max_{c_{rk}} \quad \left(\sum_{r \in \mathcal{R}} c_{rk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \qquad \text{s.t.} \quad w_n = \sum_{r \in \mathcal{R}} c_{rk} p_{rk}.$$

Using the aggregate of goods originating from r consumed by the worker in k over all possible shipping transport modes, $c_{rk} = \left[\sum_{l \in \mathcal{M}} \left(c_{rk}^{l} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}$, in Equation (A.1), the worker's maximisation problem is given by

$$\max_{c_{rk}} \quad \left(\sum_{r \in \mathcal{R}} c_{rk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \qquad \text{s.t.} \quad w_n = \sum_{r \in \mathcal{R}} c_{rk} p_{rk}.$$

The Lagrangian and resulting two first-order conditions are

$$\mathcal{L} = \left(\sum_{r \in \mathcal{R}} c_{rk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} - \lambda \left(\sum_{r \in \mathcal{R}} c_{rk} p_{rk} - w_n \right), \tag{A.2}$$

$$\frac{\partial \mathcal{L}}{\partial c_{rk}} = 0 \quad \Leftrightarrow \quad \left(\sum_{r \in \mathcal{R}} c_{rk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} c_{rk}^{-\frac{1}{\sigma}} = \lambda p_{rk},$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \quad \Leftrightarrow \quad w_n = \sum_{r \in \mathcal{R}} c_{rk} p_{rk}.$$

From the first first-order condition, for any two regions r and r',

$$\frac{c_{rk}^{-\frac{1}{\sigma}}}{c_{r'k}^{-\frac{1}{\sigma}}} = \frac{p_{rk}}{p_{r'k}}.$$

Re-arranging and summing over all regions r', I obtain

$$\begin{aligned}
& c_{r'k}^{\frac{1}{\sigma}} p_{r'k} = c_{rk}^{\frac{1}{\sigma}} p_{rk} \\
\Leftrightarrow \quad & c_{r'k} p_{r'k} = c_{rk} p_{rk}^{\sigma} p_{r'k}^{1-\sigma} \\
\Leftrightarrow \quad & \sum_{r' \in \mathcal{R}} c_{r'k} p_{r'k} = c_{rk} p_{rk}^{\sigma} \sum_{r' \in \mathcal{R}} \left(p_{r'k}\right)^{1-\sigma}.
\end{aligned}$$

Using the second first-order condition (budget constraint) and the price aggregate for goods consumed in region k, $P_k^{1-\sigma} = \sum_{r' \in \mathcal{R}} \left(p_{r'k}\right)^{1-\sigma}$, I obtain the worker's consumption of goods from region r by

$$\begin{aligned}
& w_n = c_{rk} p_{rk}^{\sigma} P_k^{1-\sigma} \\
\Leftrightarrow \quad & c_{rk} = \frac{p_{rk}^{-\sigma}}{P_k^{1-\sigma}} w_n. \qquad \text{(A.3)}
\end{aligned}$$

The value of the worker's demand is price times quantity,

$$c_{rk} p_{rk} = \left(\frac{p_{rk}}{P_k}\right)^{1-\sigma} w_n.$$

The price of delivered goods p_{rk} is the factory-gate price in origin region r multiplied with transport costs, $p_{rk} = p_r t_{rk}$, such that the trade flow from region r to the worker living in region k and working in region n is given by:

$$c_{rk} p_r t_{rk} = \left(\frac{p_r t_{rk}}{P_k}\right)^{1-\sigma} w_n. \qquad \text{(A.4)}$$

Note that transport costs are an index of transport costs over all modes.

Shipping Mode Decision

For each origin region, the worker decides on the quantity of consumption goods shipped via each transport mode. The maximisation problem is given by

$$\max_{c_{rk}^l} \quad \left(\sum_{l\in\mathcal{M}} \left(c_{rk}^l\right)^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} \qquad \text{s.t.} \quad \sum_{l\in\mathcal{M}} c_{rk}^l p_{rk}^l = c_{rk} p_{rk}.$$

The Lagrangian and the resulting two first-order conditions are

$$\begin{aligned}
\mathcal{L} &= \left(\sum_{l\in\mathcal{M}} \left(c_{rk}^l\right)^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} - \lambda\left(\sum_{l\in\mathcal{M}} c_{rk}^l p_{rk}^l - c_{rk}p_{rk}\right), \\
\frac{\partial\mathcal{L}}{\partial c_{rk}^l} = 0 \quad &\Leftrightarrow \quad \left(\sum_{l\in\mathcal{M}} \left(c_{rk}^l\right)^{\frac{\rho-1}{\rho}}\right)^{\frac{1}{\rho-1}} \left(c_{rk}^l\right)^{-\frac{1}{\rho}} = \lambda p_{rk}^l, \\
\frac{\partial\mathcal{L}}{\partial\lambda} = 0 \quad &\Leftrightarrow \quad c_{rk}p_{rk} = \sum_{l\in\mathcal{M}} c_{rk}^l p_{rk}^l.
\end{aligned}$$

From the first first-order condition, for any two modes l and l',

$$\frac{\left(c_{rk}^l\right)^{-\frac{1}{\rho}}}{\left(c_{rk}^{l'}\right)^{-\frac{1}{\rho}}} = \frac{p_{rk}^l}{p_{rk}^{l'}}.$$

Re-arranging and summing over all modes l', I obtain

$$\sum_{l'\in\mathcal{M}} c_{rk}^{l'} p_{rk}^{l'} = c_{rk}^l \left(p_{rk}^l\right)^{\rho} \sum_{l'\in\mathcal{M}} \left(p_{rk}^{l'}\right)^{1-\rho}.$$

Using the second first-order condition (budget constraint), I obtain

$$\begin{aligned}
c_{rk}p_{rk} &= c_{rk}^l \left(p_{rk}^l\right)^{\rho} \sum_{l'\in\mathcal{M}} \left(p_{rk}^{l'}\right)^{1-\rho}, \\
\Leftrightarrow \qquad c_{rk}^l &= \frac{\left(p_{rk}^l\right)^{-\rho}}{\sum_{l'\in\mathcal{M}} \left(p_{rk}^{l'}\right)^{1-\rho}} c_{rk}p_{rk}. \qquad \text{(A.5)}
\end{aligned}$$

The value of the worker's demand for goods from region r shipped via mode l is price times quantity,

$$c_{rk}^{l} p_{rk}^{l} = \frac{\left(p_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(p_{rk}^{l'}\right)^{1-\rho}} c_{rk} p_{rk}.$$

The price of delivered goods p_{rk}^{l} is the factory-gate price in origin region r multiplied with iceberg transport costs for mode l, t_{rk}^{l}.

$$c_{rk}^{l} p_r t_{rk}^{l} = \frac{\left(p_r t_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(p_r t_{rk}^{l'}\right)^{1-\rho}} c_{rk} p_{rk}. \tag{A.6}$$

Defining the price index for goods from r consumed in k, $p_{rk}^{1-\rho} = \sum_{l' \in \mathcal{M}} \left(p_{rk}^{l'}\right)^{1-\rho}$, I obtain:

$$\begin{aligned} \left(p_{rk}\right)^{1-\rho} &= \sum_{l' \in \mathcal{M}} \left(p_r t_{rk}^{l'}\right)^{1-\rho} \\ \Leftrightarrow \qquad p_{rk} &= p_r \left[\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}} = p_r t_{rk}, \end{aligned}$$

where the transport cost index is defined by

$$t_{rk} = \left[\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}. \tag{A.7}$$

The aggregate price index for goods consumed in region k can hence be expressed by

$$\begin{aligned} P_k^{1-\sigma} &= \sum_{r' \in \mathcal{R}} \left(p_{r'k}\right)^{1-\sigma} \\ &= \sum_{r' \in \mathcal{R}} \left(p_{r'} t_{r'k}\right)^{1-\sigma} \\ &= \sum_{r' \in \mathcal{R}} \left(p_{r'} \left[\sum_{l' \in \mathcal{M}} \left(t_{r'k}^{l'}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}\right)^{1-\sigma}. \end{aligned}$$

A.2 Derivations for Location and Commuting Choices

As a prerequisite, I derive indirect utility, where I replace the expression for consumption from Equation (3) into Equation (1):

$$\begin{aligned}
U_{kn}^{m*}(\omega) =& \left[\sum_{r\in\mathcal{R}}\left(\frac{p_{rk}^{-\sigma}}{P_k^{1-\sigma}}w_n\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}\\
=&\frac{w_n}{P_k^{1-\sigma}}\left[\sum_{r\in\mathcal{R}}\left(p_{rk}^{-\sigma}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}\\
=&\frac{w_n}{P_k^{1-\sigma}}\left[\sum_{r\in\mathcal{R}}p_{rk}^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}\\
=&\frac{w_n}{P_k^{1-\sigma}}\left[\sum_{r\in\mathcal{R}}\left(p_{rk}\right)^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}\\
=&\frac{w_n}{P_k^{1-\sigma}}\left[P_k^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}\\
=&\frac{w_n}{P_k}\frac{b_{kn}(\omega)v^m(\omega)}{d_{kn}^m}.
\end{aligned}\tag{A.8}$$

Worker mobility is modelled as a nested discrete choice problem. In the upper nest workers choose a residence-workplace pair, while in the lower nest they choose the transport mode for commuting between the locations. For their choice of a residence-workplace pair, workers draw $b_{kn}(\omega)$ independently from a Fréchet distribution with shape parameter $\theta > 1$ (the location choice elasticity), cumulative distribution function $F(a) = \exp\left(-a^{-\theta}\right)$, and density function $f(a) = \theta a^{-1-\theta}\exp\left(-a^{-\theta}\right)$. After choosing a residence-workplace pair, workers draw $v^m(\omega)$ independently from a Fréchet distribution with shape parameter $\eta > 1$ (the commuting mode choice elasticity) and choose a transport mode for commuting from their residence to their workplace.[30]

Commuting Mode Choice

The probability that a worker chooses mode m, conditional on having chosen to

[30] For ease of notation, I am omitting ω from the expression. So, in the following, v^m is used to denote $v^m(\omega)$ and b_{kn} is used to denote $b_{kn}(\omega)$.

commute from k to n and realising her preferences v^m, is

$$\begin{aligned} Prob_{m|kn} =& Prob\left(\frac{w_n}{P_k}\frac{b_{kn}v^m}{d_{kn}^m} \geq \frac{w_n}{P_k}\frac{b_{kn}v^{m'}}{d_{kn}^{m'}} \quad \forall \quad m' \neq m\right) \\ =& Prob\left(\frac{v^m}{d_{kn}^m} \geq \frac{v^{m'}}{d_{kn}^{m'}} \quad \forall \quad m' \neq m\right) \\ =& Prob\left(v^{m'} \leq \frac{v^m}{d_{kn}^m} d_{kn}^{m'} \quad \forall \quad m' \neq m\right) \end{aligned}$$

I derive the choice probability as in Herzog (2024) and according to Train (2009). If v^m is considered as given, the expression is the cumulative distribution function for each $v^{m'}$ evaluated at $\frac{v^m}{d_{kn}^m} d_{kn}^{m'}$. With the v's being independent, the cumulative distribution over all $m' \neq m$ is the product of the individual cumulative distributions:

$$Prob_{m|kn,v^m} = \prod_{m' \neq m} \exp\left(-\left(\frac{v^m}{d_{kn}^m} d_{kn}^{m'}\right)^{-\eta}\right).$$

As v^m is not given, the choice probability is the integral of $Prob_{m|kn,v^m}$ over all values of v^m weighted by its density:

$$\begin{aligned} Prob_{m|kn} &= \int_0^\infty \prod_{m' \neq m} \exp\left(-\left(\frac{v^m}{d_{kn}^m} d_{kn}^{m'}\right)^{-\eta}\right) \eta \left(v^m\right)^{-1-\eta} \exp\left(-\left(v^m\right)^{-\eta}\right) dv^m \\ &= \int_0^\infty \prod_{m'} \exp\left(-\left(\frac{v^m}{d_{kn}^m} d_{kn}^{m'}\right)^{-\eta}\right) \eta \left(v^m\right)^{-1-\eta} dv^m \\ &= \int_0^\infty \exp\left(-\left(\frac{v^m}{d_{kn}^m}\right)^{-\eta} \sum_{m'} \left(d_{kn}^{m'}\right)^{-\eta}\right) \eta \left(v^m\right)^{-1-\eta} dv^m. \end{aligned} \tag{A.9}$$

Define

$$h = -\left(\frac{v^m}{d_{kn}^m}\right)^{-\eta} \sum_{m'} \left(d_{kn}^{m'}\right)^{-\eta}, \tag{A.10}$$

such that

$$\begin{aligned} dh =& \eta \left(\frac{v^m}{d_{kn}^m} \right)^{-\eta-1} \frac{1}{d_{kn}^m} \sum_{m'} \left(d_{kn}^{m'} \right)^{-\eta} dv^m \\ =& \eta \left(v^m \right)^{-\eta-1} \left(\frac{1}{d_{kn}^m} \right)^{-\eta} \sum_{m'} \left(d_{kn}^{m'} \right)^{-\eta} dv^m, \end{aligned}$$

which can be re-arranged to

$$\eta \left(v^m \right)^{-1-\eta} dv^m = \frac{\left(d_{kn}^m \right)^{-\eta}}{\sum_{m'} \left(d_{kn}^{m'} \right)^{-\eta}} dh. \tag{A.11}$$

Substitute for (A.10) and (A.11) in Equation (A.9) and note that in (A.10), as $v^m \to \infty$, $h \to 0$ and as $v^m \to 0$, $h \to -\infty$, such that the bounds change. I obtain the probability of a worker choosing commuting mode m, given that she commutes from k to n:

$$\begin{aligned} Prob_{m|kn} =& \int_{-\infty}^{0} \exp\left(h\right) \frac{\left(d_{kn}^m \right)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'} \right)^{-\eta}} dh \\ =& \frac{\left(d_{kn}^m \right)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'} \right)^{-\eta}}. \end{aligned} \tag{A.12}$$

Location Choice

The probability that worker ω chooses residence-workplace pair kn, where $n \in \mathcal{R}(k)$, after realising her preferences b_{kn} is

$$\begin{aligned} Prob_{kn} =& Prob \left(\frac{w_n}{P_k \overline{d_{kn}}} b_{kn} \geq \frac{w_s}{P_r \overline{d_{rs}}} b_{rs} \quad \forall \quad rs \neq kn, \text{ with } r \in \mathcal{R},\, s \in \mathcal{R}(r) \right) \\ =& Prob \left(b_{rs} \leq \frac{w_n P_r \overline{d_{rs}}}{w_s P_k \overline{d_{kn}}} b_{kn} \quad \forall \quad rs \neq kn, \text{ with } r \in \mathcal{R},\, s \in \mathcal{R}(r) \right) \end{aligned}$$

I derive the choice probability analogously to the commuting mode choice above. If b_{kn} is considered as given, the expression is the cumulative distribution function for each b_{rs} evaluated at $\frac{w_n P_r \overline{d_{rs}}}{w_s P_k \overline{d_{kn}}} b_{kn}$. With the b's being independent, the cumulative

distribution over all $rs \neq kn$ is the product of the individual cumulative distributions:

$$Prob_{kn|b_{kn}} = \prod_{rs \neq kn} \exp\left(-\left(\frac{w_n P_r \overline{d_{rs}}}{w_s P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta}\right).$$

As b_{kn} is not given, the choice probability is the integral of $Prob_{kn|b_{kn}}$ over all values of b_{kn} weighted by its density:

$$\begin{aligned} Prob_{kn} &= \int_0^\infty \prod_{rs \neq kn} \exp\left(-\left(\frac{w_n P_r \overline{d_{rs}}}{w_s P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta}\right) \theta b_{kn}^{-1-\theta} \exp\left(-b_{kn}^{-\theta}\right) db_{kn} \\ &= \int_0^\infty \prod_{rs} \exp\left(-\left(\frac{w_n P_r \overline{d_{rs}}}{w_s P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta}\right) \theta b_{kn}^{-1-\theta} db_{kn} \\ &= \int_0^\infty \exp\left(-\left(\frac{w_n}{P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta} \sum_{rs} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta}\right) \theta b_{kn}^{-1-\theta} db_{kn}. \end{aligned} \tag{A.13}$$

Define

$$z = -\left(\frac{w_n}{P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta} \sum_{rs} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta}, \tag{A.14}$$

such that

$$\begin{aligned} dz =& \theta \left(\frac{w_n}{P_k \overline{d_{kn}}} b_{kn}\right)^{-\theta-1} \frac{w_n}{P_k \overline{d_{kn}}} \sum_{rs} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta} db_{kn} \\ =& \theta \left(\frac{w_n}{P_k \overline{d_{kn}}}\right)^{-\theta} b_{kn}^{-\theta-1} \sum_{rs} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta} db_{kn}, \end{aligned}$$

which can be re-arranged to

$$\theta b_{kn}^{-1-\theta} db_{kn} = \frac{\left(\frac{P_k \overline{d_{kn}}}{w_n}\right)^{-\theta}}{\sum_{rs} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta}} dz. \tag{A.15}$$

Substitute for (A.14) and (A.15) in Equation (A.13) and note that in (A.14), as $b_{kn} \to \infty$, $z \to 0$ and as $b_{kn} \to 0$, $z \to -\infty$, such that the bounds change. I obtain

the probability of a worker choosing to live in k and work in n:

$$Prob_{kn} = \int_{-\infty}^{0} \exp(z) \frac{\left(\frac{P_k \overline{d_{kn}}}{w_n}\right)^{-\theta}}{\sum_{r\in\mathcal{R}} \sum_{s\in\mathcal{R}} \left(\frac{P_r \overline{d_{rs}}}{w_s}\right)^{-\theta}} dz$$
$$= \frac{\left(\frac{w_n}{P_k \overline{d_{kn}}}\right)^{\theta}}{\sum_{r\in\mathcal{R}} \sum_{s\in\mathcal{R}} \left(\frac{w_s}{P_r \overline{d_{rs}}}\right)^{\theta}}. \tag{A.16}$$

A.3 Derivations for Structural Gravity System of Trade Flows

For deriving the structural gravity system of regional trade, following Anderson and van Wincoop (2003), re-arrange the goods market clearing condition (13) and divide by aggregate output of all regions, $Y = \sum_{r\in\mathcal{R}} Y_r$, yielding

$$(p_r)^{1-\sigma} = \frac{Y_r/Y}{\sum_{k\in\mathcal{R}} \left(\frac{t_{rk}}{P_k}\right)^{1-\sigma} E_k/Y}.$$

Define $\Pi_r^{1-\sigma} \equiv \sum_{k\in\mathcal{R}} \left(\frac{t_{rk}}{P_k}\right)^{1-\sigma} \frac{E_k}{Y}$ to obtain:

$$(p_r)^{1-\sigma} = \frac{Y_r/Y}{\Pi_r^{1-\sigma}}.$$

Substituting this into Equation (9) yields the structural gravity system:

$$X_{rk} = \frac{Y_r E_k}{Y} \left(\frac{t_{rk}}{\Pi_r P_k}\right)^{1-\sigma}, \tag{A.17}$$

where the outward and inward multilateral resistance terms are given by:

$$\Pi_r^{1-\sigma} = \sum_{k\in\mathcal{R}} \left(\frac{t_{rk}}{P_k}\right)^{1-\sigma} \frac{E_k}{Y} \tag{A.18}$$

$$P_k^{1-\sigma} = \sum_{r\in\mathcal{R}} \left(\frac{t_{rk}}{\Pi_r}\right)^{1-\sigma} \frac{Y_r}{Y}. \tag{A.19}$$

To obtain a gravity equation of trade by transport mode, combine trade shipped

via mode l from region r to region k from Equation (11) with the regional gravity equation (A.17):

$$X_{rk}^{l} = \frac{\left(t_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}} \frac{Y_r E_k}{Y} \left(\frac{\left[\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}}{\Pi_r P_k} \right)^{1-\sigma}.$$

Combining transport cost terms yields:

$$X_{rk}^{l} = \frac{Y_r E_k}{Y} \left(\frac{1}{\Pi_r P_k} \right)^{1-\sigma} \frac{\left(t_{rk}^{l}\right)^{1-\rho}}{\left[\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}\right]^{\frac{\sigma-\rho}{1-\rho}}}. \tag{A.20}$$

A.4 Bilateral Trade Flows With Trade Imbalances

Regional trade imbalances allow regional expenditure on goods exceeding (or below) residents' labour income and are defined as: $D_k = E_k - \sum_{n \in \mathcal{R}(k)} w_n L_{kn}$. Due to the geographic separation of work and consumption, regional trade imbalances are not as easily identified in the bilateral trade matrix as in a model without commuting, where trade imbalances are the differences between total regional exports (regional output) and total regional imports (regional expenditure). Nonetheless, if observed in the data, they can be accounted for in the model. To do so, the budget constraint of the worker living in region k and working in region n from (2) needs to allow spending beyond or below her wage:

$$w_n + \Delta_k = \sum_{r \in \mathcal{R}} c_{rk} p_{rk}.$$

Assuming all residents in k have the same surplus or deficit spending Δ_k, the aggregate trade deficit of region k is $D_k = \sum_{n \in \mathcal{R}(k)} L_{kn} \Delta_k = L_k^R \Delta_k$.

The trade flow from region r to the worker living in region k and working in

region n from Equation (3) is given by

$$c_{rk} p_r t_{rk} = \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} (w_n + \Delta_k) \,.$$

The trade flow from region r to all workers living in k and working in n is obtained by multiplying with L_{kn}:

$$c_{rk} p_r t_{rk} L_{kn} = \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} (w_n + \Delta_k) \, L_{kn}.$$

Summing over all workplaces I obtain

$$\begin{aligned} c_{rk} p_r t_{rk} L_k^R &= \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} \sum_{n \in \mathcal{R}(k)} (w_n + \Delta_k) \, L_{kn} \\ &= \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} \left(\sum_{n \in \mathcal{R}(k)} w_n L_{kn} + \sum_{n \in \mathcal{R}(k)} \Delta_k L_{kn} \right) \\ &= \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} \left(\sum_{n \in \mathcal{R}(k)} w_n L_{kn} + D_k \right). \end{aligned}$$

With $X_{rk} = c_{rk} p_r t_{rk} L_k^R$ and using the definition of trade imbalances, $E_k = \sum_{n \in \mathcal{R}(k)} w_n L_{kn} + D_k$, I obtain bilateral trade flows from region r to region k, expressed in the same way as in Equation (9), but in this case accounting for trade imbalances:

$$X_{rk} = \left(\frac{p_r t_{rk}}{P_k} \right)^{1-\sigma} E_k,$$

where

$$E_k = \sum_{n \in \mathcal{R}(k)} w_n L_{kn} + D_k.$$

Finally, note that trade imbalances in the data for a region k are not simply observed as the difference between total exports and total imports, but are obtained by the difference between total expenditure of region k on goods from all origin regions (i.e.

total imports) and the total income of all residents in region k:

$$D_k = \sum_{n \in \mathcal{R}(k)} X_{nk} - \sum_{n \in \mathcal{R}(k)} w_n L_{kn}.$$

A.5 Commute Cost Index

The commute cost index $\overline{d_{kn}}$ is the inverse of the expected utility contribution of the optimal mode choice for commuting between k and n and affects workers' residence-workplace choices, alongside wages in the workplace region and consumer prices in the residence region. To determine the commute cost index, note that the cumulative distribution function and expected value of a Fréchet distributed random variable X are given by:

$$\begin{aligned} F(x) =& Prob(X \leq x) = \exp\left(-\left(\frac{x}{s}\right)^{-\eta}\right), \\ E[X] =& s\Gamma\left(1 - \frac{1}{\eta}\right), \end{aligned} \tag{A.21}$$

where $\Gamma(\cdot)$ is the Gamma function, $s > 0$ is a scale parameter, and $\eta > 1$ is a shape parameter.[31] To obtain an expression for the commute cost index, I calculate the expected value of $\max_m \frac{v^m}{d_{kn}^m}$ conditional on kn, as in Herzog (2024). The cumulative distribution function of the random variable $\max_m \frac{v^m}{d_{kn}^m}$ evaluated at some constant $\frac{1}{d}$ can be expressed by

$$\begin{aligned} F_{kn}\left(\frac{1}{d}\right) =& Prob\left(\max_m \frac{v^m}{d_{kn}^m} \leq \frac{1}{d}\right) \\ =& Prob\left(\frac{v^m}{d_{kn}^m} \leq \frac{1}{d} \quad \forall \quad m\right) \\ =& Prob\left(v^m \leq \frac{d_{kn}^m}{d} \quad \forall \quad m\right) \end{aligned}$$

[31] The expected value of the Fréchet distributed random variable X is only finite when the shape parameter $\eta > 1$. For expected utility to be finite requires $\eta > 1$.

For $\frac{d_{kn}^m}{d}$ given, this expression is the cumulative distribution function for each Fréchet-distributed v^m evaluated at $\frac{d_{kn}^m}{d}$. With the v's being independent, the cumulative distribution over all m is the product of the individual cumulative distributions:

$$\begin{aligned} F_{kn}\left(\frac{1}{d}\right) &= \prod_m \exp\left(-\left(\frac{d_{kn}^m}{d}\right)^{-\eta}\right) \\ &= \exp\left(-\left(\frac{1}{d}\right)^{-\eta}\sum_m (d_{kn}^m)^{-\eta}\right) \\ &= \exp\left(-\left[\frac{\frac{1}{d}}{\left(\sum_m (d_{kn}^m)^{-\eta}\right)^{\frac{1}{\eta}}}\right]^{-\eta}\right) \end{aligned}$$

This expression corresponds with the cumulative distribution function of a Fréchet distributed random variable in the above specification (A.21), where $x = \frac{1}{d}$ and $s = \left(\sum_m (d_{kn}^m)^{-\eta}\right)^{\frac{1}{\eta}}$. It allows me to write down the expected value of $\max_m \frac{v^m}{d_{kn}^m}$, conditional on location pair kn:

$$E\left[\max_m \frac{v^m}{d_{kn}^m}\bigg|_{kn}\right] = \left(\sum_m (d_{kn}^m)^{-\eta}\right)^{\frac{1}{\eta}} \Gamma\left(1-\frac{1}{\eta}\right).$$

With $\frac{1}{\overline{d_{kn}}}$ defined as the expected value of $\max_m \frac{v^m}{d_{kn}^m}$ for the given residence-workplace pair kn, the commute cost index is given by

$$\overline{d_{kn}} = E\left[\max_m \frac{v^m}{d_{kn}^m}\bigg|_{kn}\right]^{-1} = \left(\sum_m (d_{kn}^m)^{-\eta}\right)^{-\frac{1}{\eta}} \left[\Gamma\left(1-\frac{1}{\eta}\right)\right]^{-1}. \tag{A.22}$$

A.6 Model Equations in Levels

The general equilibrium of regional trade, regional commuting, as well as transport mode choices for trade and commuting is given by the following set of equations, taking into account that perfect competition implies wages equal to producer prices $w_n = p_n$. Trade imbalances D_k are treated as exogenous, since they are observed in the data.[32]

[32] Details on trade imbalances are given in Appendix A.4.

$$\pi_{rk} = \frac{(w_r t_{rk})^{1-\sigma}}{P_k^{1-\sigma}} \tag{A.23}$$

$$X_{rk} = \pi_{rk} E_k \tag{A.24}$$

$$Y_r = \sum_{k \in \mathcal{R}} X_{rk} \tag{A.25}$$

$$w_n = \frac{Y_n}{\sum_{k \in \mathcal{R}(n)} L_{kn}} \tag{A.26}$$

$$E_k = \sum_{n \in \mathcal{R}(k)} w_n L_{kn} + D_k \tag{A.27}$$

$$P_k^{1-\sigma} = \sum_{r \in \mathcal{R}} (w_r t_{rk})^{1-\sigma} \tag{A.28}$$

$$X_{rk}^l = \frac{\left(t_{rk}^l\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'}\right)^{1-\rho}} X_{rk} \tag{A.29}$$

$$t_{rk} = \left[\sum_{l \in \mathcal{M}} \left(t_{rk}^l\right)^{1-\rho}\right]^{\frac{1}{1-\rho}} \tag{A.30}$$

$$L_{kn} = \frac{\left(\frac{w_n}{P_k \overline{d_{kn}}}\right)^{\theta}}{\sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{R}(r)} \left(\frac{w_s}{P_r \overline{d_{rs}}}\right)^{\theta}} \overline{L} \tag{A.31}$$

$$L_{kn}^m = \frac{\left(d_{kn}^m\right)^{-\eta}}{\sum_{m' \in \mathcal{M}} \left(d_{kn}^{m'}\right)^{-\eta}} L_{kn} \tag{A.32}$$

$$\overline{d_{kn}} = \left(\sum_{m \in \mathcal{M}} \left(d_{kn}^m\right)^{-\eta}\right)^{-\frac{1}{\eta}} \left[\Gamma\left(1 - \frac{1}{\eta}\right)\right]^{-1} \tag{A.33}$$

A.7 Derivations of the General Equilibrium in Changes: Equations (22)–(32)

Starting from the system of equations (A.23)–(A.33), I derive the General Equilibrium in changes, given in the system of equations (22)–(32). Let a hat above a variable denote the variable's counterfactual value relative to its baseline value, e.g. $\widehat{\pi}_{rk} = \frac{\pi_{rk}^c}{\pi_{rk}^b}$. I assume that the total labour force, $\overline{L}$, is constant. Trade imbalances D_k are treated as exogenous and constant.

$$
\begin{aligned}
\widehat{P}_k^{1-\sigma} =& \frac{\sum_{r\in\mathcal{R}} \left(w_r^c t_{rk}^c\right)^{1-\sigma}}{\sum_{r\in\mathcal{R}} \left(w_r^b t_{rk}^b\right)^{1-\sigma}} \\
=& \frac{\sum_{r\in\mathcal{R}} \left(w_r^c t_{rk}^c\right)^{1-\sigma} \frac{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}}{\sum_{j\in\mathcal{R}} \left(w_j^b t_{jk}^b\right)^{1-\sigma}} \\
=& \frac{\sum_{r\in\mathcal{R}} \left(w_r^b t_{rk}^b\right)^{1-\sigma} \left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\sum_{j\in\mathcal{R}} \left(w_j^b t_{jk}^b\right)^{1-\sigma}} \\
=& \sum_{r\in\mathcal{R}} \frac{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}{\sum_{j\in\mathcal{R}} \left(w_j^b t_{jk}^b\right)^{1-\sigma}} \left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma} \\
=& \sum_{r\in\mathcal{R}} \pi_{rk}^b \left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma} \\
\Leftrightarrow \widehat{P}_k =& \left(\sum_{r\in\mathcal{R}} \pi_{rk}^b \left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}
\end{aligned}
$$

$$
\begin{aligned}
\widehat{\pi}_{rk} =& \frac{\frac{\left(w_r^c t_{rk}^c\right)^{1-\sigma}}{\left(P_k^c\right)^{1-\sigma}}}{\frac{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}{\left(P_k^b\right)^{1-\sigma}}} = \frac{\frac{\left(w_r^c t_{rk}^c\right)^{1-\sigma}}{\sum_{l\in\mathcal{R}}\left(w_l^c t_{lk}^c\right)^{1-\sigma}}}{\frac{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}{\sum_{l\in\mathcal{R}}\left(w_l^b t_{lk}^b\right)^{1-\sigma}}} = \frac{\frac{\left(w_r^c t_{rk}^c\right)^{1-\sigma}}{\left(w_r^b t_{rk}^b\right)^{1-\sigma}}}{\frac{\sum_{l\in\mathcal{R}}\left(w_l^c t_{lk}^c\right)^{1-\sigma}}{\sum_{l\in\mathcal{R}}\left(w_l^b t_{lk}^b\right)^{1-\sigma}}} \\
=& \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\frac{\sum_{l\in\mathcal{R}}\left(w_l^c t_{lk}^c\right)^{1-\sigma} \frac{\left(w_l^b t_{lk}^b\right)^{1-\sigma}}{\left(w_l^b t_{lk}^b\right)^{1-\sigma}}}{\sum_{l\in\mathcal{R}}\left(w_l^b t_{lk}^b\right)^{1-\sigma}}} \\
=& \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\frac{\sum_{l\in\mathcal{R}}\left(w_l^b t_{lk}^b\right)^{1-\sigma}\left(\widehat{w}_l \widehat{t}_{lk}\right)^{1-\sigma}}{\sum_{j\in\mathcal{R}}\left(w_j^b t_{jk}^b\right)^{1-\sigma}}} \\
=& \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\sum_{l\in\mathcal{R}} \frac{\left(w_l^b t_{lk}^b\right)^{1-\sigma}}{\sum_{j\in\mathcal{R}}\left(w_j^b t_{jk}^b\right)^{1-\sigma}} \left(\widehat{w}_l \widehat{t}_{lk}\right)^{1-\sigma}} \\
=& \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\sum_{l\in\mathcal{R}} \pi_{lk}^b \left(\widehat{w}_l \widehat{t}_{lk}\right)^{1-\sigma}} \\
=& \frac{\left(\widehat{w}_r \widehat{t}_{rk}\right)^{1-\sigma}}{\widehat{P}_k^{1-\sigma}}
\end{aligned}
$$

$$\widehat{X}_{rk} = \frac{\pi^c_{rk} E^c_k}{\pi^b_{rk} E^b_k} = \widehat{\pi}_{rk} \widehat{E}_k$$

$$\begin{aligned}
\widehat{Y}_r Y^b_r = & Y^c_r = \sum_{k \in \mathcal{R}} X^c_{rk} \\
= & \sum_{k \in \mathcal{R}} \widehat{X}_{rk} X^b_{rk} \\
= & \sum_{k \in \mathcal{R}} \widehat{\pi}_{rk} \widehat{E}_k \pi^b_{rk} E^b_k
\end{aligned}$$

$$\begin{aligned}
\widehat{w}_n = & \frac{\frac{Y^c_n}{\sum_{k \in \mathcal{R}(n)} L^c_{kn}}}{\frac{Y^b_n}{\sum_{k \in \mathcal{R}(n)} L^b_{kn}}} = \frac{\frac{Y^c_n}{Y^b_n}}{\frac{\sum_{k \in \mathcal{R}(n)} L^c_{kn}}{\sum_{k \in \mathcal{R}(n)} L^b_{kn}}} = \frac{\widehat{Y}_n}{\frac{\sum_{k \in \mathcal{R}(n)} L^c_{kn} \frac{L^b_{kn}}{L^b_{kn}}}{\sum_{k \in \mathcal{R}(n)} L^b_{kn}}} \\
= & \frac{\widehat{Y}_n}{\frac{\sum_{k \in \mathcal{R}(n)} L^b_{kn} \widehat{L}_{kn}}{\sum_{j \in \mathcal{R}(n)} L^b_{jn}}} \\
= & \frac{\widehat{Y}_n}{\sum_{k \in \mathcal{R}(n)} \frac{L^b_{kn}}{\sum_{j \in \mathcal{R}(n)} L^b_{jn}} \widehat{L}_{kn}}
\end{aligned}$$

$$\begin{aligned}
\widehat{E}_k E^b_k = & E^c_k = \sum_{n \in \mathcal{R}(k)} w^c_n L^c_{kn} + D_k \\
= & \sum_{n \in \mathcal{R}(k)} \widehat{w}_n w^b_n \widehat{L}_{kn} L^b_{kn} + D_k \\
= & \sum_{n \in \mathcal{R}(k)} \widehat{w}_n \frac{Y^b_n}{\sum_{j \in \mathcal{R}(n)} L^b_{jn}} \widehat{L}_{kn} L^b_{kn} + D_k
\end{aligned}$$

$$
\begin{aligned}
\widehat{X}_{rk}^{l} =& \frac{\frac{\left(t_{rk}^{l,c}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',c}\right)^{1-\rho}} X_{rk}^{c}}{\frac{\left(t_{rk}^{l,b}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',b}\right)^{1-\rho}} X_{rk}^{b}} = \frac{\frac{\left(t_{rk}^{l,c}\right)^{1-\rho}}{\left(t_{rk}^{l,b}\right)^{1-\rho}}}{\frac{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',c}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',b}\right)^{1-\rho}}} \widehat{X}_{rk} \\
=& \frac{\left(\widehat{t}_{rk}^{l}\right)^{1-\rho}}{\frac{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',c}\right)^{1-\rho} \frac{\left(t_{rk}^{l',b}\right)^{1-\rho}}{\left(t_{rk}^{l',b}\right)^{1-\rho}}}{\sum_{j \in \mathcal{M}} \left(t_{rk}^{j,b}\right)^{1-\rho}}} \widehat{X}_{rk} \\
=& \frac{\left(\widehat{t}_{rk}^{l}\right)^{1-\rho}}{\frac{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l',b}\right)^{1-\rho} \left(\widehat{t}_{rk}^{l'}\right)^{1-\rho}}{\sum_{j \in \mathcal{M}} \left(t_{rk}^{j,b}\right)^{1-\rho}}} \widehat{X}_{rk} \\
=& \frac{\left(\widehat{t}_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \frac{\left(t_{rk}^{l',b}\right)^{1-\rho}}{\sum_{j \in \mathcal{M}} \left(t_{rk}^{j,b}\right)^{1-\rho}} \left(\widehat{t}_{rk}^{l'}\right)^{1-\rho}} \widehat{X}_{rk} \\
=& \frac{\left(\widehat{t}_{rk}^{l}\right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \pi_{rk}^{l',b} \left(\widehat{t}_{rk}^{l'}\right)^{1-\rho}} \widehat{X}_{rk}
\end{aligned}
$$

$$
\begin{aligned}
\widehat{t}_{rk} =& \frac{\left[\sum_{l \in \mathcal{M}} \left(t_{rk}^{l,c}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}}{\left[\sum_{l \in \mathcal{M}} \left(t_{rk}^{l,b}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}} = \left[\frac{\sum_{l \in \mathcal{M}} \left(t_{rk}^{l,c}\right)^{1-\rho}}{\sum_{l \in \mathcal{M}} \left(t_{rk}^{l,b}\right)^{1-\rho}}\right]^{\frac{1}{1-\rho}} = \left[\frac{\sum_{l \in \mathcal{M}} \left(t_{rk}^{l,b}\right)^{1-\rho} \left(\widehat{t}_{rk}^{l}\right)^{1-\rho}}{\sum_{j \in \mathcal{M}} \left(t_{rk}^{j,b}\right)^{1-\rho}}\right]^{\frac{1}{1-\rho}} \\
=& \left[\sum_{l \in \mathcal{M}} \frac{\left(t_{rk}^{l,b}\right)^{1-\rho}}{\sum_{j \in \mathcal{M}} \left(t_{rk}^{j,b}\right)^{1-\rho}} \left(\widehat{t}_{rk}^{l}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}} \\
=& \left[\sum_{l \in \mathcal{M}} \pi_{rk}^{l,b} \left(\widehat{t}_{rk}^{l}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}
\end{aligned}
$$

$$\widehat{L}_{kn} = \frac{\frac{\left(\frac{w_n^c}{P_k^c \widehat{d_{kn}^c}}\right)^\theta}{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\left(\frac{w_s^c}{P_r^c d_{rs}^c}\right)^\theta}\overline{L}}{\frac{\left(\frac{w_n^b}{P_k^b \widehat{d_{kn}^b}}\right)^\theta}{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\left(\frac{w_s^b}{P_r^b d_{rs}^b}\right)^\theta}\overline{L}} = \frac{\frac{\left(\frac{w_n^c}{P_k^c \widehat{d_{kn}^c}}\right)^\theta}{\left(\frac{w_n^b}{P_k^b \widehat{d_{kn}^b}}\right)^\theta}}{\frac{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\left(\frac{w_s^c}{P_r^c d_{rs}^c}\right)^\theta}{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\left(\frac{w_s^b}{P_r^b d_{rs}^b}\right)^\theta}} = \frac{\left(\frac{\widehat{w}_n}{\widehat{P}_k \widehat{d_{kn}}}\right)^\theta}{\frac{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\left(\frac{w_s^b}{P_r^b d_{rs}^b}\right)^\theta\left(\frac{\widehat{w}_s}{\widehat{P}_r \widehat{d_{rs}}}\right)^\theta}{\sum_{i\in\mathcal{R}}\sum_{j\in\mathcal{R}(i)}\left(\frac{w_j^b}{P_i^b d_{ij}^b}\right)^\theta}}$$

$$= \frac{\left(\frac{\widehat{w}_n}{\widehat{P}_k \widehat{d_{kn}}}\right)^\theta}{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\frac{\left(\frac{w_s^b}{P_r^b d_{rs}^b}\right)^\theta}{\sum_{i\in\mathcal{R}}\sum_{j\in\mathcal{R}(i)}\left(\frac{w_j^b}{P_i^b d_{ij}^b}\right)^\theta}\left(\frac{\widehat{w}_s}{\widehat{P}_r \widehat{d_{rs}}}\right)^\theta}$$

$$= \frac{\left(\frac{\widehat{w}_n}{\widehat{P}_k \widehat{d_{kn}}}\right)^\theta}{\sum_{r\in\mathcal{R}}\sum_{s\in\mathcal{R}(r)}\frac{L_{rs}^b}{\overline{L}}\left(\frac{\widehat{w}_s}{\widehat{P}_r \widehat{d_{rs}}}\right)^\theta}$$

$$\widehat{d_{kn}} = \frac{\left(\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,c}\right)^{-\eta}\right)^{-\frac{1}{\eta}}\left[\Gamma\left(1-\frac{1}{\eta}\right)\right]^{-1}}{\left(\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,b}\right)^{-\eta}\right)^{-\frac{1}{\eta}}\left[\Gamma\left(1-\frac{1}{\eta}\right)\right]^{-1}} = \frac{\left(\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,c}\right)^{-\eta}\right)^{-\frac{1}{\eta}}}{\left(\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,b}\right)^{-\eta}\right)^{-\frac{1}{\eta}}}$$

$$= \left(\frac{\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,b}\right)^{-\eta}\left(\widehat{d}_{kn}^{m}\right)^{-\eta}}{\sum_{m\in\mathcal{M}}\left(d_{kn}^{m,b}\right)^{-\eta}}\right)^{-\frac{1}{\eta}}$$

$$= \left(\sum_{m\in\mathcal{M}}\frac{\left(d_{kn}^{m,b}\right)^{-\eta}}{\sum_{j\in\mathcal{M}}\left(d_{kn}^{j,b}\right)^{-\eta}}\left(\widehat{d}_{kn}^{m}\right)^{-\eta}\right)^{-\frac{1}{\eta}}$$

$$= \left(\sum_{m\in\mathcal{M}}\frac{L_{kn}^{m,b}}{L_{kn}^{b}}\left(\widehat{d}_{kn}^{m}\right)^{-\eta}\right)^{-\frac{1}{\eta}}$$

$$\Leftrightarrow \left(\widehat{d_{kn}}\right)^{-\eta} = \sum_{m\in\mathcal{M}}\frac{L_{kn}^{m,b}}{L_{kn}^{b}}\left(\widehat{d}_{kn}^{m}\right)^{-\eta}$$

$$
\begin{aligned}
\widehat{L}_{kn}^{m} =& \frac{\frac{\left(d_{kn}^{m,c}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',c}\right)^{-\eta}} L_{kn}^{c}}{\frac{\left(d_{kn}^{m,b}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',b}\right)^{-\eta}} L_{kn}^{b}} = \frac{\frac{\left(d_{kn}^{m,c}\right)^{-\eta}}{\left(d_{kn}^{m,b}\right)^{-\eta}} \frac{L_{kn}^{c}}{L_{kn}^{b}}}{\frac{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',c}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',b}\right)^{-\eta}}} = \frac{\left(\widehat{d}_{kn}^{m}\right)^{-\eta} \widehat{L}_{kn}}{\frac{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',c}\right)^{-\eta}}{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',b}\right)^{-\eta}}} \\
=& \frac{\left(\widehat{d}_{kn}^{m}\right)^{-\eta} \widehat{L}_{kn}}{\sum_{m' \in \mathcal{M}}\left(d_{kn}^{m',c}\right)^{-\eta} \frac{\left(d_{kn}^{m',b}\right)^{-\eta}}{\sum_{j \in \mathcal{M}}\left(d_{kn}^{j,b}\right)^{-\eta}}} \\
=& \frac{\left(\widehat{d}_{kn}^{m}\right)^{-\eta} \widehat{L}_{kn}}{\sum_{m' \in \mathcal{M}} \frac{\left(d_{kn}^{m',b}\right)^{-\eta}}{\sum_{j \in \mathcal{M}}\left(d_{kn}^{j,b}\right)^{-\eta}} \left(\widehat{d}_{kn}^{m'}\right)^{-\eta}} \\
=& \frac{\left(\widehat{d}_{kn}^{m}\right)^{-\eta} \widehat{L}_{kn}}{\sum_{m' \in \mathcal{M}} \frac{L_{kn}^{m',b}}{L_{kn}^{b}} \left(\widehat{d}_{kn}^{m'}\right)^{-\eta}} \\
=& \frac{\left(\widehat{d}_{kn}^{m}\right)^{-\eta} \widehat{L}_{kn}}{\left(\widehat{\overline{d_{kn}}}\right)^{-\eta}} \\
=& \frac{\left(\widehat{\overline{d_{kn}}}\right)^{\eta} \widehat{L}_{kn}}{\left(\widehat{d}_{kn}^{m}\right)^{\eta}}
\end{aligned}
$$

A.8 Expected Utility

Expected utility is the expected value of the indirect utility in Equation (6). Conditional on a residence-workplace pair kn, workers choose the commuting mode m that maximises utility. The utility associated with living in region k and working in region n can then be written as

$$
U_{kn}^{*}(\omega) = \frac{w_n}{P_k} b_{kn}(\omega) \max_{m} \frac{v^m(\omega)}{d_{kn}^{m}}.
$$

Replacing the lower nest commuting mode utility component by its expected value, $E\left[\max_m \frac{v^m}{d_{kn}^m}\middle|_{kn}\right] = \frac{1}{\overline{d_{kn}}} = \left(\sum_m \left(d_{kn}^m\right)^{-\eta}\right)^{\frac{1}{\eta}} \Gamma\left(1-\frac{1}{\eta}\right)$ from Equation (A.22), the util-

ity relevant for the choice of residence-workplace pair kn in the upper nest is

$$\tilde{U}^*_{kn}(\omega) = \frac{w_n}{P_k\overline{d_{kn}}} b_{kn}(\omega),$$

where $\frac{w_n}{P_k\overline{d_{kn}}}$ is treated as deterministic, leaving $b_{kn}(\omega)$ as the only idiosyncratic variation. Workers choose the residence-workplace pair kn which maximises their utility, such that

$$\tilde{U}^*(\omega) = \max_{kn} \tilde{U}^*_{kn}(\omega) = \max_{kn} \frac{w_n}{P_k\overline{d_{kn}}} b_{kn}(\omega).$$

The cumulative distribution function of the random variable $\max_{kn} \frac{w_n}{P_k\overline{d_{kn}}} b_{kn}(\omega)$ evaluated at some arbitrary constant u can be expressed by

$$\begin{aligned} F(u) =& Prob\left(\max_{kn} \frac{w_n}{P_k\overline{d_{kn}}} b_{kn} \leq u\right) \\ =& Prob\left(\frac{w_n}{P_k\overline{d_{kn}}} b_{kn} \leq u \quad \forall \quad kn\right) \\ =& Prob\left(b_{kn} \leq \frac{u}{\frac{w_n}{P_k\overline{d_{kn}}}} \quad \forall \quad kn\right). \end{aligned}$$

For $\frac{u}{\frac{w_n}{P_k\overline{d_{kn}}}}$ given, this expression is the cumulative distribution function for each Fréchet-distributed b_{kn} evaluated at $\frac{u}{\frac{w_n}{P_k\overline{d_{kn}}}}$. With the b's being independent, the cumulative distribution over all kn is the product of the individual cumulative distributions:

$$\begin{aligned} F(u) =& \prod_{kn} \exp\left[-\left(\frac{u}{\frac{w_n}{P_k\overline{d_{kn}}}}\right)^{-\theta}\right] \\ =& \prod_{kn} \exp\left[-u^{-\theta}\left(\frac{w_n}{P_k\overline{d_{kn}}}\right)^{\theta}\right] \\ =& \exp\left[-u^{-\theta}\sum_{kn}\left(\frac{w_n}{P_k\overline{d_{kn}}}\right)^{\theta}\right] \end{aligned}$$

$$= \exp\left[-\left(\frac{u}{\left(\sum_{kn}\left(\frac{w_n}{P_k\overline{\overline{d_{kn}}}}\right)^{\theta}\right)^{\frac{1}{\theta}}}\right)^{-\theta}\right]$$

This expression corresponds with the cumulative distribution function of a Fréchet distributed random variable, with scale $\left(\sum_{kn}\left(\frac{w_n}{P_k\overline{\overline{d_{kn}}}}\right)^{\theta}\right)^{\frac{1}{\theta}}$ and shape θ. It allows me to write down the expected value of $\max_{kn}\frac{w_n}{P_k\overline{\overline{d_{kn}}}}b_{kn}$, i.e. expected utility:

$$\begin{aligned}
E\left[\tilde{U}^*\right] =& E\left[\max_{kn}\tilde{U}^*_{kn}\right] = E\left[\max_{kn}\frac{w_n}{P_k\overline{\overline{d_{kn}}}}b_{kn}\right] \\
=& \left(\sum_{kn}\left(\frac{w_n}{P_k\overline{\overline{d_{kn}}}}\right)^{\theta}\right)^{\frac{1}{\theta}}\Gamma\left(1-\frac{1}{\theta}\right) \\
=& \left(\sum_{kn}\left(\frac{w_n}{P_k}\right)^{\theta}\left(\frac{1}{\overline{\overline{d_{kn}}}}\right)^{\theta}\right)^{\frac{1}{\theta}}\Gamma\left(1-\frac{1}{\theta}\right) \\
=& \left(\sum_{kn}\left(\frac{w_n}{P_k}\right)^{\theta}\left(\left(\sum_{m}(d^m_{kn})^{-\eta}\right)^{\frac{1}{\eta}}\Gamma\left(1-\frac{1}{\eta}\right)\right)^{\theta}\right)^{\frac{1}{\theta}}\Gamma\left(1-\frac{1}{\theta}\right) \\
=& \Gamma\left(1-\frac{1}{\theta}\right)\Gamma\left(1-\frac{1}{\eta}\right)\left(\sum_{kn}\left(\frac{w_n}{P_k}\right)^{\theta}\left(\sum_{m}(d^m_{kn})^{-\eta}\right)^{\frac{\theta}{\eta}}\right)^{\frac{1}{\theta}}.
\end{aligned} \tag{A.34}$$

For expected utility to be finite, the location choice elasticity must satisfy $\theta > 1$ and the commuting mode choice elasticity must satisfy $\eta > 1$.

Expected utility is the expected maximum utility of any worker before realising the residence-workplace preference shocks b_{kn} and the commuting-mode preference shocks v^m. Ex ante, expected utility for all workers is the same. Ex post, different workers have different welfare due to different idiosyncratic shocks.

The change in expected utility is given by

$$\begin{aligned}
\widehat{E\left[\tilde{U}^{*}\right]} =& \frac{\left(\sum_{kn}\left(\frac{w_n^c}{P_k^c\overline{d_{kn}}^c}\right)^{\theta}\right)^{\frac{1}{\theta}}}{\left(\sum_{kn}\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}\right)^{\frac{1}{\theta}}} = \left[\frac{\sum_{kn}\frac{\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}}{\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}}\left(\frac{w_n^c}{P_k^c\overline{d_{kn}}^c}\right)^{\theta}}{\sum_{kn}\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}}\right]^{\frac{1}{\theta}} \\
=& \left[\frac{\sum_{kn}\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}\left(\frac{\widehat{w}_n}{\widehat{P}_k\widehat{\overline{d_{kn}}}}\right)^{\theta}}{\sum_{rs}\left(\frac{w_s^b}{P_r^b\overline{d_{rs}}^b}\right)^{\theta}}\right]^{\frac{1}{\theta}} = \left[\sum_{kn}\frac{\left(\frac{w_n^b}{P_k^b\overline{d_{kn}}^b}\right)^{\theta}}{\sum_{rs}\left(\frac{w_s^b}{P_r^b\overline{d_{rs}}^b}\right)^{\theta}}\left(\frac{\widehat{w}_n}{\widehat{P}_k\widehat{\overline{d_{kn}}}}\right)^{\theta}\right]^{\frac{1}{\theta}} \\
=& \left[\sum_{kn}\frac{L_{kn}^b}{\overline{L}^b}\left(\frac{\widehat{w}_n}{\widehat{P}_k\widehat{\overline{d_{kn}}}}\right)^{\theta}\right]^{\frac{1}{\theta}}. \qquad \text{(A.35)}
\end{aligned}$$

B Discrete Choice Model of Goods Transport Mode

The share of trade from r to k shipped via mode l in total trade from r to k, derived from the nested CES demand structure, can alternatively be derived as transport mode choice probability using a discrete choice model. In this model, the worker's utility function is given by

$$U_{kn}^{m}(\omega) = \left[\sum_{r\in\mathcal{R}} \left(c_{rk}^{l}\epsilon_{rk}^{l}(\omega)\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \frac{b_{kn}(\omega)v^{m}(\omega)}{d_{kn}^{m}},$$

where $\epsilon_{rk}^{l}(\omega)$ are worker-specific idiosyncratic transport-mode preference shocks drawn independently from a Fréchet distribution with shape parameter $\phi > 1$ (the transport mode choice elasticity). The shocks are drawn after the decision on an origin region r where to source goods from, upon which the worker chooses via which transport mode l to ship the goods to her region of residence k.

The optimal shipping mode choice depends on the utility obtained with it. Therefore, consider a given origin r and transport mode l. The worker in k then maximises utility subject to the budget constraint of spending at most the value spent on goods from the origin region, $c_{rk}^{l}p_{rk}^{l} = c_{rk}p_{rk}$, yielding consumption of

$$c_{rk}^{l} = \frac{c_{rk}p_{rk}}{p_{rk}^{l}}.$$

The price of delivered goods p_{rk}^{l} is the factory-gate price in origin region r multiplied with iceberg transport costs for mode l, t_{rk}^{l}, $p_{rk}^{l} = p_{r}t_{rk}^{l}$. Indirect utility for the worker in region k from consuming goods from region r shipped via mode l is

$$U_{rk}^{l*}(\omega) = \frac{c_{rk}p_{rk}}{p_{r}t_{rk}^{l}}\epsilon_{rk}^{l}(\omega).$$

For a given origin r, the worker in k chooses a shipping mode l if the utility

obtained using mode l is at least as high as the utility obtained using any other mode l' among the set of available transport modes $\mathcal{M}$. The probability that worker ω chooses mode l, conditional on having chosen to source goods from r to k and realising her preferences ϵ^l_{rk},[33] is

$$\begin{aligned} Prob_{l|rk} =& Prob\left(\frac{c_{rk}p_{rk}}{p_r t^l_{rk}}\epsilon^l_{rk} \geq \frac{c_{rk}p_{rk}}{p_r t^{l'}_{rk}}\epsilon^{l'}_{rk} \quad \forall \quad l' \neq l\right) \\ =& Prob\left(\frac{\epsilon^l_{rk}}{t^l_{rk}} \geq \frac{\epsilon^{l'}_{rk}}{t^{l'}_{rk}} \quad \forall \quad l' \neq l\right) \\ =& Prob\left(\epsilon^{l'}_{rk} \leq \frac{\epsilon^l_{rk}}{t^l_{rk}} t^{l'}_{rk} \quad \forall \quad l' \neq l\right). \end{aligned}$$

I derive the choice probability as in McFadden (1974) and according to Train (2009). If ϵ^l_{rk} is considered given and with the ϵ's being independent, the probability conditional on ϵ^l_{rk} can be expressed as

$$Prob_{l|rk,\epsilon^l_{rk}} = \prod_{l' \neq l} \exp\left(-\left(\frac{\epsilon^l_{rk}}{t^l_{rk}} t^{l'}_{rk}\right)^{-\phi}\right)$$

As ϵ^l_{rk} is not given, the choice probability is the integral of $Prob_{l|rk,\epsilon^l_{rk}}$ over all values of ϵ^l_{rk} weighted by its density,

$$\begin{aligned} Prob_{l|rk} &= \int_0^\infty \prod_{l' \neq l} \exp\left(-\left(\frac{\epsilon^l_{rk}}{t^l_{rk}} t^{l'}_{rk}\right)^{-\phi}\right) \phi \left(\epsilon^l_{rk}\right)^{-1-\phi} \exp\left(-\left(\epsilon^l_{rk}\right)^{-\phi}\right) d\epsilon^l_{rk} \\ &= \int_0^\infty \prod_{l'} \exp\left(-\left(\frac{\epsilon^l_{rk}}{t^l_{rk}} t^{l'}_{rk}\right)^{-\phi}\right) \phi \left(\epsilon^l_{rk}\right)^{-1-\phi} d\epsilon^l_{rk} \\ &= \int_0^\infty \exp\left(-\left(\frac{\epsilon^l_{rk}}{t^l_{rk}}\right)^{-\phi} \sum_{l'} \left(t^{l'}_{rk}\right)^{-\phi}\right) \phi \left(\epsilon^l_{rk}\right)^{-1-\phi} d\epsilon^l_{rk}. \end{aligned} \tag{B.1}$$

Define

$$g = -\left(\frac{\epsilon^l_{rk}}{t^l_{rk}}\right)^{-\phi} \sum_{l'} \left(t^{l'}_{rk}\right)^{-\phi}, \tag{B.2}$$

[33] For ease of notation, I am omitting ω from the expression. So, in the following, ϵ^l_{rk} is used to denote $\epsilon^l_{rk}(\omega)$.

such that

$$\begin{aligned} dg =& \phi \left(\frac{\epsilon_{rk}^{l}}{t_{rk}^{l}} \right)^{-\phi-1} \frac{1}{t_{rk}^{l}} \sum_{l'} \left(t_{rk}^{l'} \right)^{-\phi} d\epsilon_{rk}^{l} \\ =& \phi \left(\epsilon_{rk}^{l} \right)^{-\phi-1} \left(\frac{1}{t_{rk}^{l}} \right)^{-\phi} \sum_{l'} \left(t_{rk}^{l'} \right)^{-\phi} d\epsilon_{rk}^{l}, \end{aligned}$$

which can be re-arranged to

$$\phi \left(\epsilon_{rk}^{l} \right)^{-1-\phi} d\epsilon_{rk}^{l} = \frac{\left(t_{rk}^{l} \right)^{-\phi}}{\sum_{l'} \left(t_{rk}^{l'} \right)^{-\phi}} dg. \tag{B.3}$$

Substitute for (B.2) and (B.3) in Equation (B.1) and note that in (B.2), as $\epsilon_{rk}^{l} \to \infty$, $g \to 0$ and as $\epsilon_{rk}^{l} \to 0$, $g \to -\infty$, such that the bounds change. I obtain the probability of a worker choosing shipping mode l, given that she sources goods from r to k:

$$\begin{aligned} Prob_{l|rk} =& \int_{-\infty}^{0} \exp\left(g\right) \frac{\left(t_{rk}^{l} \right)^{-\phi}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'} \right)^{-\phi}} dg \\ =& \frac{\left(t_{rk}^{l} \right)^{-\phi}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'} \right)^{-\phi}}. \end{aligned} \tag{B.4}$$

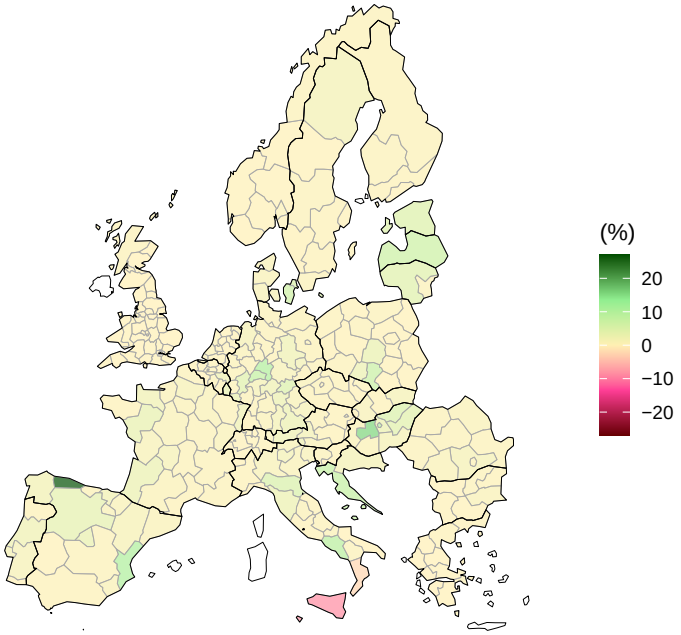
The above probability coincides with the share of goods shipped from r to k using mode l. Defining $\phi = \rho - 1$, I obtain:

$$\frac{X_{rk}^{l}}{X_{rk}} = \frac{\left(t_{rk}^{l} \right)^{1-\rho}}{\sum_{l' \in \mathcal{M}} \left(t_{rk}^{l'} \right)^{1-\rho}}, \tag{B.5}$$

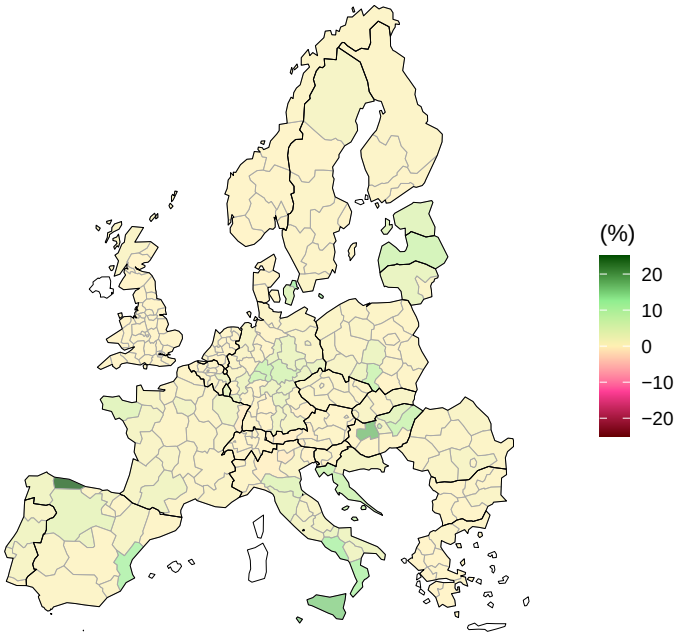
which coincides with the share in Equation (12), derived using the nested CES demand structure for shipping mode choices.

C Additional GE Results

Figure C.1: GE Results of 2013 TEN-T rail projects: Change in total exports and imports (aggregated over transport modes).



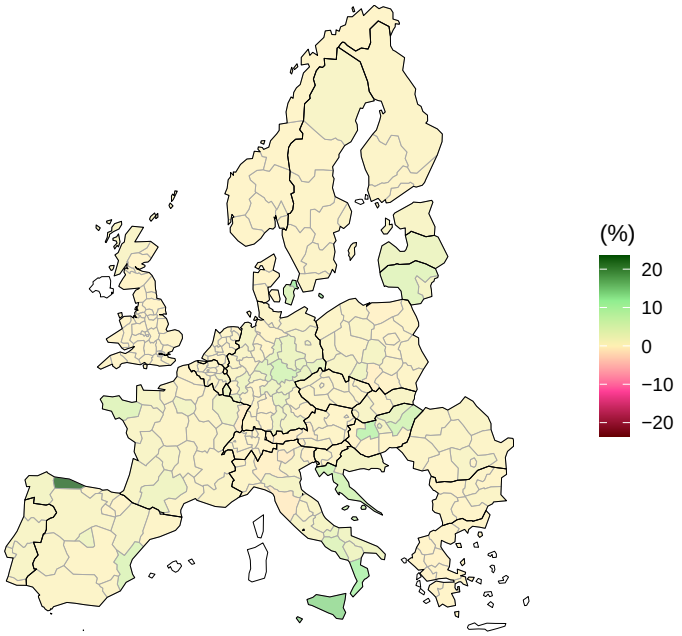
(a) Exports change.



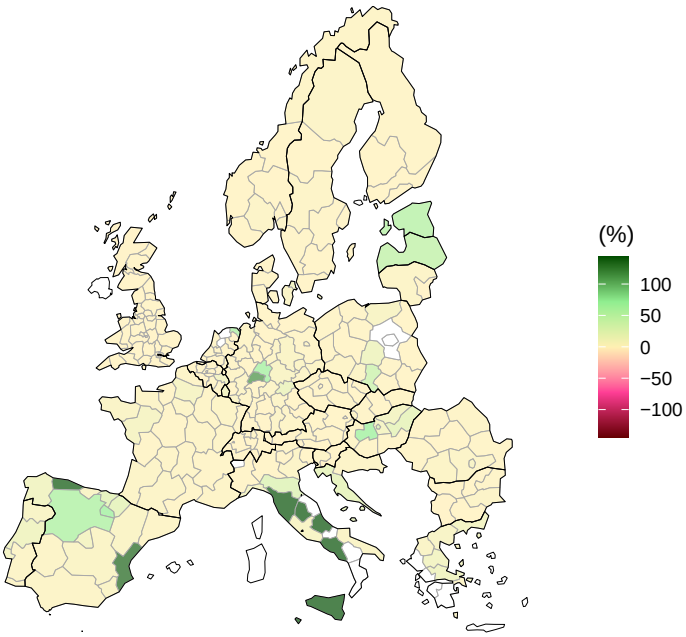
(b) Imports change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure C.2: GE Results of 2013 TEN-T rail projects: Change in imports via road and rail.



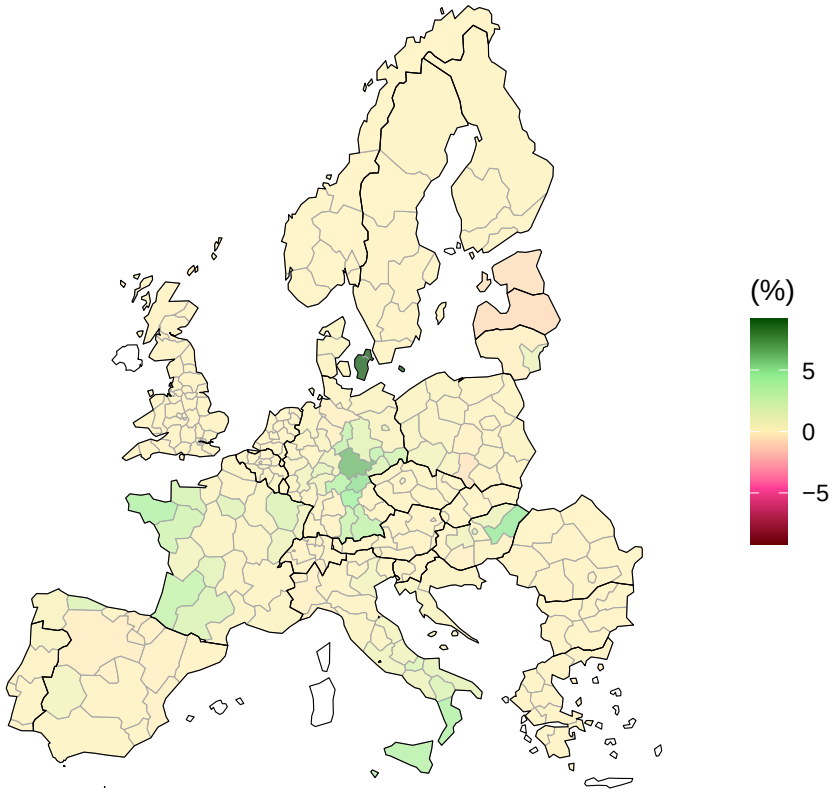
(a) Road imports change.



(b) Rail imports change.

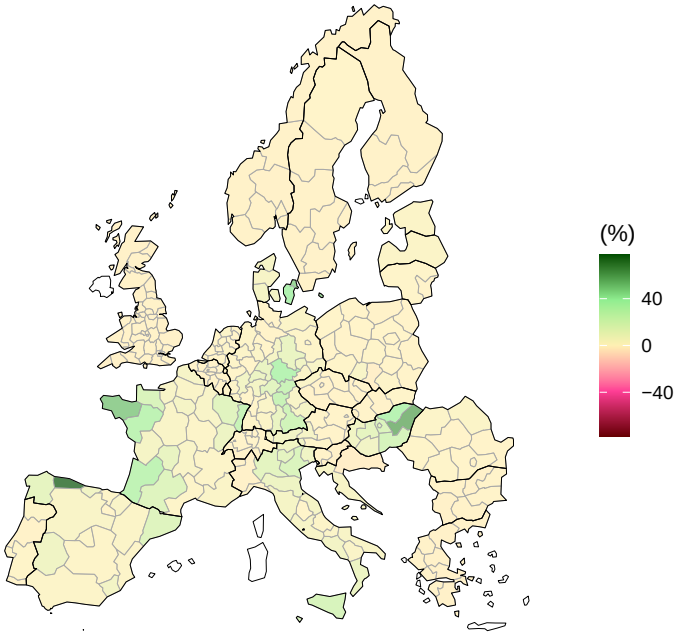
Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.
Panel (b): For the following regions, the change in rail imports is not coloured to scale: Italian region Umbria (ITI2, 267.89%), Italian region Toscana (ITI1, 269.38%), Italian region Campania (ITF3, 316.36%), Italian region Sicilia (ITG1, 484.16%).

Figure C.3: GE Results of 2013 TEN-T rail projects: Change in domestic sales.

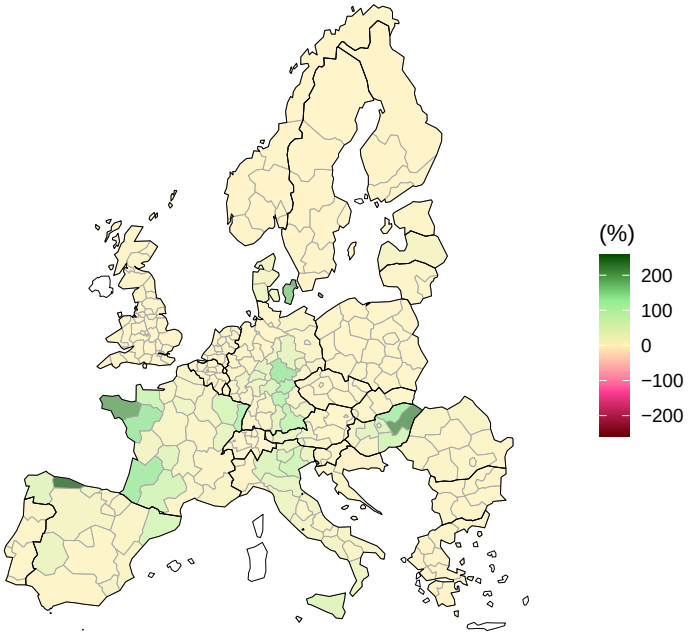


Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data. For the Danish capital region Hovedstaden (DK01), the change in domestic sales of 15.89% is not coloured to scale.

Figure C.4: GE Results of 2013 TEN-T rail projects: Change in incommuters via road and rail.



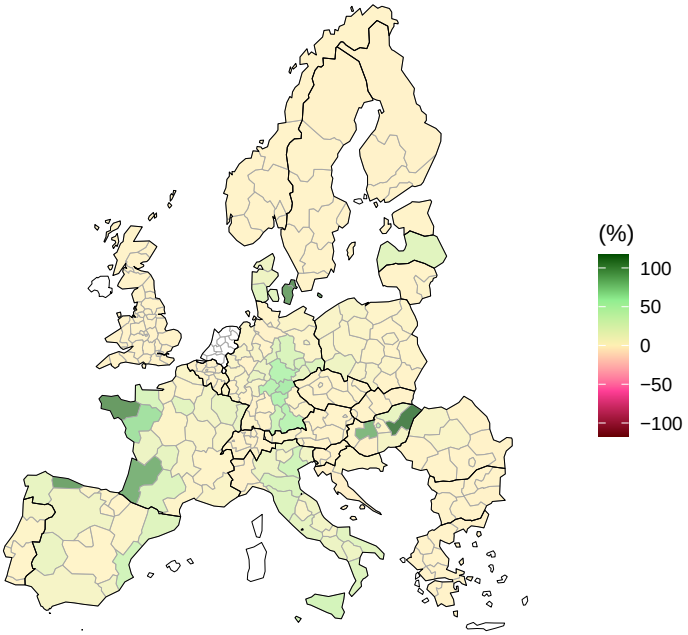
(a) Incommuters via road change.



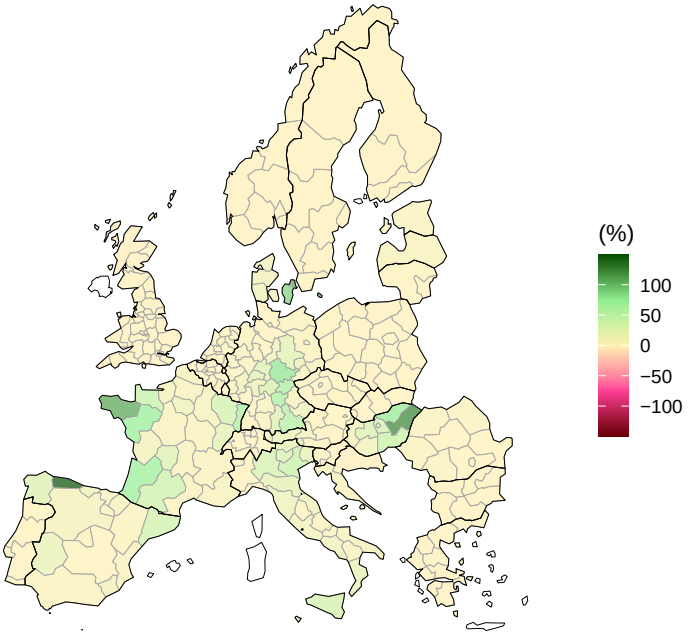
(b) Incommuters via rail change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure C.5: GE Results of 2013 TEN-T rail projects: Change in outcommuters and incommuters (aggregated over transport modes).



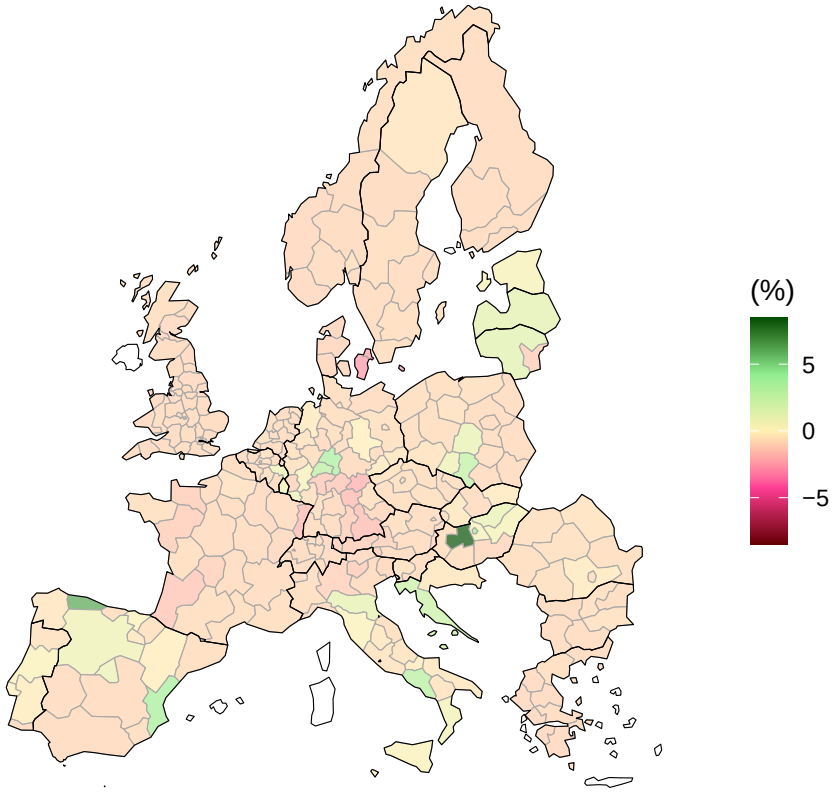
(a) Outcommuters change.



(b) Incommuters change.

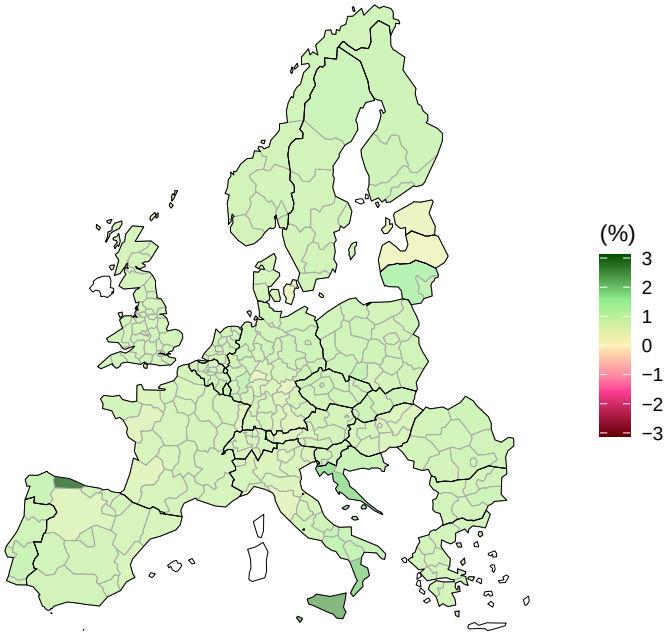
Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure C.6: GE Results of 2013 TEN-T rail projects: Change in domestic workers.

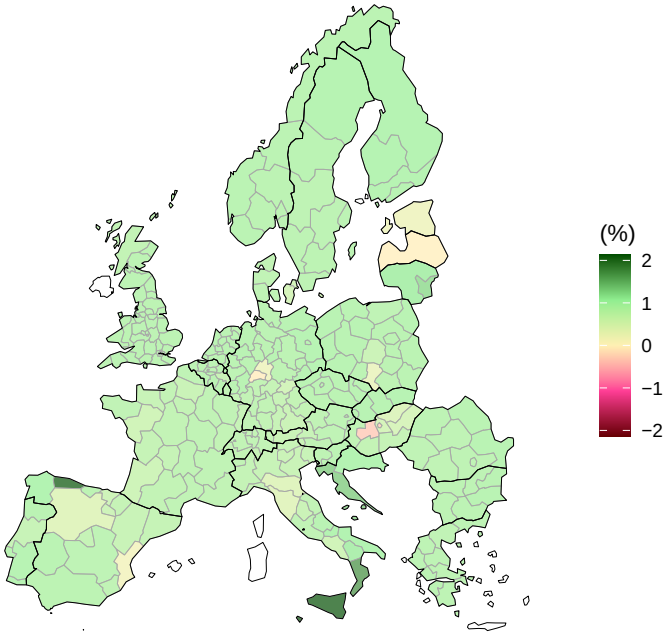


Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure C.7: GE Results of 2013 TEN-T rail projects: Change in wages and consumer prices.



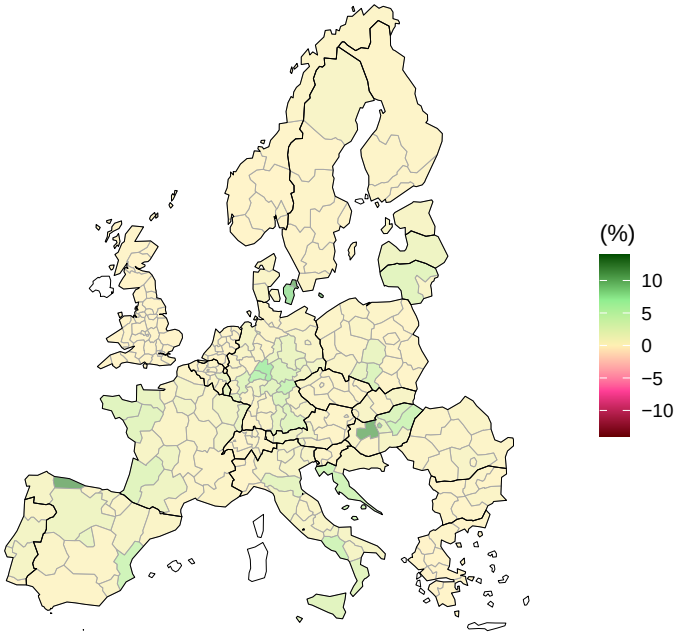
(a) Wages change.



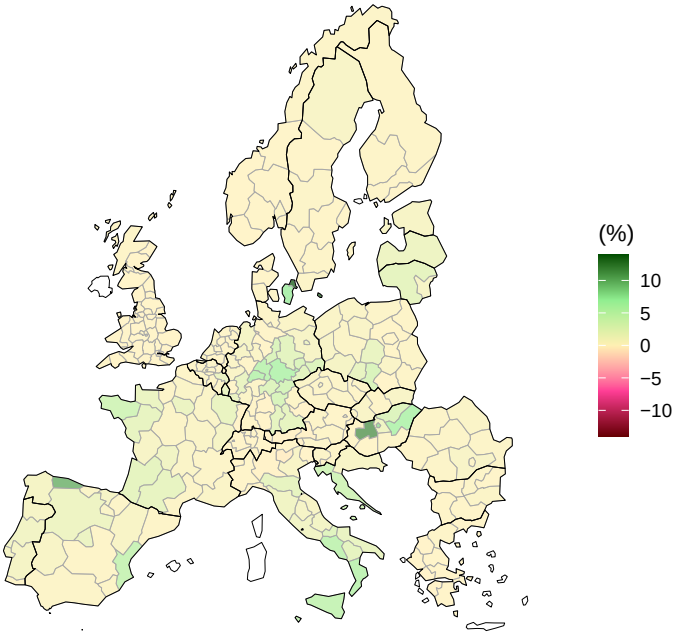
(b) Consumer prices change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.

Figure C.8: GE Results of 2013 TEN-T rail projects: Change in output and expenditure.



(a) Output change.



(b) Expenditure change.

Notes: Percentage changes for EU regions induced by completion of rail projects from the 2013 TEN-T regulation (trade and commuting cost reduction). White areas indicate regions excluded from the analysis due to missing connectivity with the TEN-T rail and road network or due to missing data.